

The Cost of Inclusion: Decomposing Bidder Exclusion in Public Procurement^{*}

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Abstract

SME set-asides are a default procurement-preference instrument worldwide. Their welfare turns on whether the price cost flows through the change in admissible bidders or through how each remaining bidder bids—a question price regressions cannot answer. I exploit a March 2018 administrative reversal that extended an SME-only rule to medical supplies on São Paulo’s centralized R\$13 billion procurement platform. A difference-in-differences against 76 never-treated product groups estimates a 10–11 percent rise in absolute winning prices, with SME participation roughly doubling. An asymmetric IPV model identified from Pregão drop-out bids decomposes the price response under observed equilibrium entry: the within-auction component accounts for 74 percent of the simulated effect in non-pharmaceuticals and 73 percent in pharmaceuticals; endogenous SME entry attenuates a 50–60 percent larger latent shock. The structural counterfactual on its own normalization is three to four times the DiD coefficient because the two objects average over related but distinct primitives. The implied welfare cost is 28.7 percent of the open-regime baseline price in non-pharmaceuticals and 47.0 percent in pharmaceuticals at $\lambda = 0.30$, scaling to a back-of-envelope R\$ 55–128 million (US\$ 16–37 million) per year of welfare loss on Group-65 procurement alone. A 10 percent SME price preference

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welfare-dominates the set-aside in thick standardized markets at near-zero fiscal cost; the comparison reverses in thin pharmaceutical markets under strict primitive invariance. The conditional ranking is itself the policy-design statement.

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JEL MSC: H32, H57, L26, L53, D44

1. Introduction

Procurement set-asides for small firms are among the most widely deployed government-procurement instruments worldwide ([Bosio et al., 2022](#)). They were designed on a textbook intuition: bringing more bidders to a protected segment should reduce procurement prices. The intuition has not survived contact with the data. A decade of empirical evidence places the price cost of bidder-exclusion rules in the ten-to-fifteen percent range ([Marion, 2007](#); [Krasnokutskaya, 2011](#); [Athey et al., 2013](#); [Nakabayashi, 2013](#); [Hyytinen et al., 2018](#)). What price regressions cannot deliver is the channel: whether the markup runs through the change in *who* can bid, or through the change in *how* each remaining bidder bids. The welfare ranking of bidder exclusion against alternative preference instruments turns on which mechanism dominates, and the policy fix differs by an order of magnitude depending on the answer. If the markup is extensive, the prescription is to widen the protected roster. If it is intensive, no roster expansion will close the gap; the only lever is to shrink the rule.

A March 2018 administrative reversal in São Paulo state, Brazil, supplies the natural experiment. The state extended an existing SME-only procurement rule to medical and hospital supplies—the last major exception—on the country’s largest centralized state-government procurement platform, the Bolsa Eletrônica de Compras (BEC). The reversal was a legal opinion internal to the state prosecutor’s office, defended on isonomy grounds

and uncorrelated with prices, competition, or supplier complaints in the affected product group; the timing was bureaucratic. A difference-in-differences against 76 never-treated product groups, reported as design validation in Appendix [Appendix I](#), estimates that absolute winning prices rose 10–11 percent on the same window used by the structural exercise, with SME participation roughly doubling. The model-implied counterfactual price effect, expressed as a ratio to the open-regime baseline, is roughly three to four times larger (~ 34 percent of p_{S_1} in non-pharma and ~ 47 percent in pharma) because it averages over a transformed counterfactual primitive (Section 7 and Appendix [Appendix I](#)); the two magnitudes measure related but distinct objects, and the welfare arithmetic of this paper is anchored to the structural object on the structural normalization. The DiD coefficient fixes the average effect on observed log prices. It does not say which mechanism produced the markup or what the rule costs in welfare terms.

Three magnitudes appear in this paper and should not be confused. The DiD coefficient (10–11 percent) is the average effect on observed log winning prices, item and month fixed effects partialled out. The structural counterfactual (+0.259 in non-pharma, +0.308 in pharma) is the simulated mean of the second-order statistic under each counterfactual entry profile, scaled by the reference price; in p_{S_1} units it is three to four times the DiD coefficient because the two objects average over different primitives. The welfare cost (28.7 percent of p_{S_1} in non-pharma, 47.0 percent in pharma at $\lambda = 0.30$) adds the marginal-cost-of-public-funds distortion on top of the allocative wedge. Sections 6 and 7 develop these in turn; Appendix [Appendix I](#) reports the DiD.

This paper builds the structure that does. I specify an asymmetric independent-private-values reading of the platform’s Pregão English-reverse auction format. Bidders draw costs from type-specific distributions F_c^{SME} and F_c^- and compete under a descending clock; losers exit at cost and the last firm supplies the contract. Drop-out prices of losers therefore point-identify F_c^k under [Haile and Tamer \(2003\)](#), without inverting a bid function

or imposing parametric structure on bidding. An auction-level common scale is shrunk out following [Krasnokutskaya \(2011\)](#); winners are treated as left-censored at the second-lowest exit and recovered by a Turnbull NPMLE robustness. Entry is handled in the [Athey et al. \(2011\)](#) spirit: pre- and post-period bidder counts are taken as equilibrium arrival rates, and the main specification allows the post-policy SME cost distribution to differ from its pre-policy counterpart as part of the equilibrium-selection object. The exercise is conducted under observed equilibrium entry; it is not a fully solved nested entry-bidding equilibrium, and the decomposition that follows is a counterfactual-simulation accounting under the modeling assumptions stated.

Three findings organize the paper.

(i) The price effect is mostly within the auction. Let p_{S_1} be the simulated winning price under the pre-policy open regime, p_{S_2} the SME-only counterfactual at the pre-policy SME pool, and p_{S_3} the SME-only counterfactual at the observed post-policy SME pool. In the main specification, $p_{S_3} - p_{S_1} = +0.259$ of the reference price in non-pharmaceuticals and $+0.308$ in pharmaceuticals (Table [D.18](#)). The within-auction piece—which I call *sheltered bidding*—is 74.5 percent of that effect in non-pharma and 73.3 percent in pharma. The share stays in the 73–85 percent range across cost-distribution estimators (Turnbull NPMLE: 74, 82) and across the two specifications of the post-policy SME pool (strict invariance: 85, 79). The set-aside’s price cost is mostly a property of how the surviving pool bids, not of how many firms it contains.

(ii) Endogenous SME entry attenuates a 50–60 percent larger latent shock. Hold the SME pool at its pre-policy size while imposing the SME-only rule and the simulated price effect rises to 152 percent of the realized one in non-pharma and 157 percent in pharma. Without endogenous entry the set-aside would cost 50–60 percent more in price and welfare. The participation margin is the government’s partial offset, not the source of the markup: the SMEs the protected segment induces to enter absorb

part of the within-auction shock, though not all of it.

(iii) A 10 percent price preference welfare-dominates the set-aside in non-pharma; the comparison bifurcates in pharma. In thick standardized non-pharma markets the preference shifts simulated prices by -0.004 against $+0.259$ under the set-aside at essentially zero fiscal cost, and welfare-dominates under both the main and the strict-invariance specifications. In thin heterogeneous pharma the preference dominates under the main equilibrium-selection reading but the ranking flips under strict invariance, with the implicit welfare weight required to prefer the set-aside falling to 0.7. The bifurcation is the diagnostic: it pins down where the modeling treatment of the protected pool is first-order, and it is exactly where the policy ranking demands caution.

Three contributions follow. First, the paper’s methodological position rests on a combination of conditions rare in the procurement-preference literature: a total-exclusion legal trigger (clean separation of regimes), a Pregão drop-out auction format (point-identification of cost distributions), an auction-level UH correction ([Krasnokutskaya, 2011](#)), an observed-equilibrium-entry treatment in the [Athey et al. \(2011\)](#) spirit, and a [Saez and Stantcheva \(2016\)](#) welfare-weight identity. Each component is established in the literature; the combination delivers a within-auction structural decomposition of an SME procurement preference where the partial-preference structural literature ([Marion, 2007](#); [Krasnokutskaya, 2011](#); [Athey et al., 2011](#)) identifies the two margins jointly with cross-equation constraints rather than from a single legal trigger. Second, it shows that endogenous SME entry is not a side issue but part of the welfare object: a fixed-pool counterfactual—the standard reduced-form shortcut—misstates the realized policy cost by roughly half. Third, it converts the welfare comparison into a market-design statement rather than a universal ranking. The 10 percent price preference welfare-dominates the set-aside in thick, standardized markets across both specifications; in thin, heterogeneous markets the comparison depends on how entry selection is modeled. The

bifurcation is the diagnostic: it identifies precisely the market conditions under which equilibrium-selection vs. primitive-invariance treatment of the protected pool is first-order, and converts a sensitivity check into a substantive claim about where bidder exclusion delivers proportional returns and where it does not. The paper sits in dialogue with a broader empirical literature on procurement design, procurement institutions, the political economy of public contracting, and the public-finance evaluation of government programs in developing economies ([Coviello and Mariniello, 2014](#); [Coviello et al., 2026](#); [Decarolis et al., 2025](#); [Colonnelli and Prem, 2022](#); [Ferraz et al., 2016](#); [Szerman, 2023](#); [Lewis-Faupel et al., 2016](#); [Olken, 2007](#); [Niehaus and Sukhtankar, 2013](#); [Best et al., 2023](#); [Bandiera et al., 2021](#); [Brogaard et al., 2021](#)), where the contribution here is methodological—bidder-level drop-out data combined with a clean total-exclusion legal trigger that permit a structural decomposition of the within-auction and participation margins on the same footing.

A pair of comparable furosemide 40 mg purchases by the same São Paulo public buyer eight months apart and bracketing the cutoff fixes the magnitudes. In February 2018 the unit cleared roughly twelve percent below the buyer’s reference price; in October 2018 it cleared just above. The difference is one realization of the average direction the design estimates across more than a hundred thousand item-transactions in the matched window of eighteen months on each side of the March 2018 cutoff—10–11 percent on the absolute final price in the difference-in-differences benchmark of Appendix [Appendix I](#), the magnitude the structural exercise of this paper interprets and prices. The headline numbers across the four reporting layers are summarized in Table [1](#).

2. Institutional Background

Brazil’s federal SME statute (Complementary Law 123/2006, subsequently amended by LC 147/2014 for procurement-rule provisions and LC 155/2016 for revenue-bracket adjustments) defines micro- and small enterprises by annual gross revenue ceilings of

Table 1: Headline magnitudes.

	Non-pharma	Pharma
DiD on $\log p^{\text{final}}$ (absolute, App. Appendix I)	+0.10 to +0.11	
DiD on $p^{\text{final}}/p^{\text{ref}}$ (App. Appendix I)	+0.06	
Simulated $p_{S_3} - p_{S_1}$, units of p^{ref}	+0.259	+0.308
Within-auction share, main specification	74.5%	73.3%
Within-auction share, Turnbull NPMLE	74.0%	82.0%
Within-auction share, strict primitive invariance	85.0%	79.0%
Welfare cost (% of p_{S_1} , $\lambda = 0.30$)	28.7%	47.0%
$w_{\star}^{\text{SME}}$ to prefer set-aside, main spec.	2.42	2.61
$w_{\star}^{\text{SME}}$, strict invariance	> 1	0.7

DiD magnitudes from the difference-in-differences benchmark in [Appendix I](#) (against 76 never-treated product groups, matched window of 18 months on each side of the March 2018 cutoff). Structural magnitudes from the asymmetric IPV counterfactual under observed equilibrium entry; sources by row: [Sec. 6](#) (decomposition), [Sec. 8](#) (Turnbull NPMLE; strict invariance), [Sec. 7](#) (welfare cost), [Sec. 9](#) (welfare weights). The pharmaceutical welfare-dominance ranking against a 10 percent price preference is robust under the main specification but reverses under strict primitive invariance ($w_{\star}^{\text{SME}} = 0.7 < 1$); the non-pharmaceutical ranking is preserved under both specifications.

R\$360,000 (US\$103,000) and R\$4,800,000 (US\$1.37 million), respectively—the latter raised from R\$3.6M to R\$4.8M effective January 2018 by LC 155/2016—and mandates exclusive SME tenders for items valued at R\$80,000 (US\$22,900) or less. Public buyer units (PBUs)—federal, state, and municipal administrative entities subject to procurement rules—can avoid the SME-only default only by justifying the departure through a formal report to the audit courts, a costly and scrutinized process. LC 147/2014 converted what had been an option (“may”) into a binding default (“must”) and imposed these *opt-out costs*; state-level adherence to SME-only tendering for eligible items rose from roughly 13% to nearly 70% in the subsequent years.¹

Since September 2000 (Decree 45.085/2000, renamed by Decree 45.695/2001), the

¹Adoption-rate figures and Group-65 sub-rates reported throughout this section are computed by the author from the BEC microdata extract used for the structural analysis ([Section 3](#)), restricted to items satisfying the federal SME-eligibility threshold of R\$80,000. The R\$4.8M EPP revenue ceiling raised by LC 155/2016 has no direct bite on the structural sample, where SME status is identified from the BEC administrative *enquadramento* flag and the Receita Federal *porte* register at the firm-auction level ([Section 3](#)); the bracket change is reported here for institutional completeness.

state of São Paulo has operated the Bolsa Eletrônica de Compras (BEC), a centralized electronic procurement platform; the Pregão electronic reverse-auction format that anchors this paper’s analytical sample was regulated within BEC by Decree 49.722/2005. BEC centralizes Pregão for standardized-goods procurement by state bureaus and affiliated units, with registered suppliers submitting sealed bids (phase 1) followed by an iterative electronic auction (phases 2 onward) in which each participant may progressively lower its offer until the auction closes.² BEC handled approximately R\$13 billion (US\$3.7 billion) of procurement in 2019 alone and has cumulatively moved more than R\$105 billion (US\$30 billion) since implementation, spanning 860,000 purchase offers and 4.8 million item-level transactions.³ BEC organizes its catalog into three tiers—group, class, and item—so that “medical, dental, and hospital equipment and supplies” form Group 65, within which class 6531 identifies medications subject to price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED).

Group 65 was a singular exception to the 2014 rule, but for a transversal legal reason rather than a sector-specific one. Until late 2017, the prevailing interpretation of art. 48, I of LC 123/2006—both at the Tribunal de Contas do Estado de São Paulo (TCE-SP) and operationally inside BEC—was that the R\$80,000 threshold for SME-only tenders applied to the *total estimated value of the procurement notice*, not to each individual item or lot. Under this restrictive reading, no notice with total estimated value above R\$80,000 could be made SME-only, regardless of whether individual items sat below the threshold. Group-65 procurement notices systematically combine small individual items

²Brazilian institutional terms used throughout the paper (Pregão, Convite, BEC, CADMAT, CMED, PGE-SP, TCE-SP, Sefaz-SP) are defined in Appendix [Appendix A](#).

³Platform institutional description at www.bec.sp.gov.br/BECSP/Quem_Somos/Quem_Somos_BEC_no_vo.aspx (retrieved April 2026). The 4.8-million item-level count and the 860,000 purchase-offer count are author tallies from the same BEC extract used in the structural analysis. **[VERIFY: 2019 monetary total of R\$13 bn and cumulative R\$105 bn to be cross-checked against the source from which they were originally drawn (Sefaz-SP annual report, BEC press release, or internal LAI response). Public BEC statistics pages return 404 as of April 2026; the author should retain the original source citation for the referee.]**

(medications, supplies, dressings) with large notice totals reflecting bulk public-sector purchasing, so the restrictive interpretation mechanically barred Group 65 from the SME-only default—not because medical supplies were classified as “strategic,” but because their typical procurement structure (small unit values, large notice totals) was exactly what the totalizing interpretation excluded. The TCE-SP Tribunal Pleno had sustained this restrictive reading by tie-breaking vote on 23 September 2015 (eTC-5509.989.15), holding that “the expression ‘items of contracting’ refers to the entirety of the object sought by the Administration, and the sum of these values cannot exceed R\$80,000.”⁴

On 11 December 2017, the Procuradoria-Geral do Estado de São Paulo (PGE-SP) issued **Parecer Sub-G Cons. nº 151/2017**, doing two things at once. First, it held that the R\$80,000 threshold applies item-by-item, allowing mixed-exclusivity procurement notices in which items below R\$80,000 are SME-only and items above remain open. Second, it ruled that the more generous federal regime introduced by LC 147/2014 supersedes the previously applicable São Paulo state acts (*Lei Estadual* 13.122/2008 and *Decreto Estadual* 54.229/2009), which had embedded a less SME-favorable interpretation in state-level procurement practice. The combined holding is what destabilized the prevailing BEC operational practice. The argument was generic—about the legal interpretation of art. 48, I and about federal-vs-state norm hierarchy—not specific to Group 65; the unlocking of Group-65 set-asides is a mechanical consequence

⁴TCE-SP, eTC-5509.989.15, Tribunal Pleno, sessão de 23/09/2015 (decisão por voto de desempate). Decisão acessível no portal de jurisprudência do TCE-SP (<https://www.tce.sp.gov.br/jurisprudencia>); a interpretação restritiva do art. 48, I LC 123/2006 que prevaleceu nesse julgamento é amplamente discutida na literatura técnico-jurídica brasileira sobre licitações para ME/EPP.

of how medical-supplies notices are structured.⁵ Operationally, BEC moved before PGE consolidated the legal interpretation: **COMUNICADO BEC nº 02/2017 (18 July 2017)** introduced the SME-only OC functionality, and **COMUNICADO BEC nº 03/2017 (20 November 2017)** expanded it to mixed-exclusivity Pregão notices combining ample-participation items, 25%-quota items, and SME-only items in the same notice—the precise operational mechanism that Parecer 151/2017 ratified three weeks later.⁶ TCE-SP did not immediately follow: under eTC-5509.989.15 the Tribunal Pleno had previously sustained the restrictive interpretation, and only reversed course on 16 May 2018 in eTC-9589.989.18, two and a half months after the empirical cutoff of this paper.⁷ The empirical cutoff of March 2018 therefore captures the moment of mass take-up of the new functionality by Group-65 PBUs, not the moment of formal regulatory enablement (which was incremental between July and November 2017); in the months after the cutoff, SME-only adherence within Group 65 jumped to 43% on average, below the rate for other groups largely because class-6531 pharmaceuticals are oligopolistic and

⁵Parecer SubG-Cons. nº 151/2017, Subprocuradoria-Geral de Consultoria da PGE-SP, ementa: “*Licitações exclusivas para ME e EPP (artigo 48, I, LC nº 123/2006) devem ser realizadas para itens ou lotes cujo valor estimado, individualmente, não ultrapasse o limite legal de R\$ 80.000,00 (oitenta mil reais), independentemente do valor total da Oferta de Compra. ... Após a reforma operada pela Lei Complementar Federal nº 147/2014, a Lei Complementar Federal nº 123/2006 passou a destinar às MEs e EPPs tratamento simplificado e diferenciado em licitações e contratações mais favorável do que aquele previsto na Lei Estadual nº 13.122/2008 e no Decreto Estadual nº 54.229/2009. Aplicabilidade, na íntegra, da normativa federal.*” The opinion was submitted to the Procurador Geral do Estado for approval, after which it became binding guidance for state-level PBUs; the parecer is officially disclosed on the BEC/SP legislation portal (https://www.bec.sp.gov.br/becsp/Legislacao/UI_Selecao.aspx, *Pareceres* section, retrieved April 2026). The Boletim CEPGE indexes only Pareceres da Procuradoria Administrativa (PA) and does not list Pareceres da Subprocuradoria-Geral de Consultoria (Sub-G Cons.); the BEC/SP legislation portal is the canonical disclosure channel for the latter.

⁶COMUNICADO BEC nº 02/2017 (18/07/2017): “*Nova Funcionalidade – geração de OC com opção de participação exclusiva ME/EPP/Cooperativa c/ Dir. Pref.*”; COMUNICADO BEC nº 03/2017 (20/11/2017): “*Nova Funcionalidade – geração de Oferta de Compra – OC – Pregão Eletrônico – Material com opções de itens – AMPLA, COTA até 25% para ME/EPP e Coop. c/ Dir. Pref. e exclusiva para ME/EPP e Coop.*” Both atos disclosed at the BEC/SP legislation portal (https://www.bec.sp.gov.br/becsp/Legislacao/UI_Selecao.aspx, seção Comunicados, retrieved April 2026).

⁷TCE-SP, eTC-9589.989.18, Tribunal Pleno, sessão de 16/05/2018, alinhando-se ao Parecer Sub-G Cons. 151/2017 da PGE-SP quanto à interpretação item-a-item do limite de R\$80 mil. Decisão acessível no portal de jurisprudência do TCE-SP (<https://www.tce.sp.gov.br/jurisprudencia>).

often fail the three-eligible-SMEs requirement.

What matters for identification is that the policy change was triggered by a transversal legalistic reinterpretation of art. 48, I at PGE-SP, not by anything happening in the Group-65 procurement market. Parecer SubG-Cons. 151/2017 made no substantive claim about medical supplies or any other product group; it pronounced on a general statutory question. The Group-65 unlocking that followed is the mechanical interaction of a generic threshold rule with the structural feature of medical procurement (small items, large notice totals). Public documentation turns up no sign that the parecer was a response to price levels, competition patterns, or supplier complaints specific to Group 65 in 2017; the timing was set by the bureaucratic process inside PGE-SP and by when BEC operationalized the new interpretation. The TCE-SP reversal in May 2018 lies after the empirical cutoff and confirms that the trigger was a top-down legal interpretation—when the audit court itself flipped, it ratified what PGE-SP had already established for state-level procurement, rather than initiating an independent change in the Group-65 environment. The change is therefore plausibly exogenous to the outcomes studied. This is the institutional basis for Assumption 3: the cutoff shifts admissibility and equilibrium entry selection, not the underlying production technology of the goods themselves. The reduced-form DiD against the 76 never-treated product groups, reported in Appendix [Appendix I](#), provides one empirical check; the primitive-invariance KS test in Table [C.9](#) provides another.

Table [2](#) consolidates the institutional chronology that anchors the empirical strategy.

3. Data and Sample Construction

The data come from BEC, the state of São Paulo’s centralized electronic procurement platform. BEC intermediates all standardized-goods procurement for state bureaus and affiliated units, so the raw extract covers the universe of item-level bids, transacted prices,

Table 2: Institutional timeline—federal and São Paulo state acts relevant to the structural

Date	Act	Content
Sep 2000	Decree 45.085/2000 (SP)	BEC platform created; renamed Decree 45.695/2001
Jun 2005	Decree 49.722/2005 (SP)	Pregão electronic reverse-auction regulated within B
Dec 2006	LC 123/2006 (federal)	SME statute (<i>Estatuto Nacional</i>).
2008	Lei Estadual 13.122/2008 (SP)	SP state-level SME procurement regulation (less favo
2009	Decree 54.229/2009 (SP)	Regulates Lei Estadual 13.122/2008.
Aug 2014	LC 147/2014 (federal)	SME-only default (“may”→“must”); R\$80k threshol
Sep 2015	TCE-SP eTC-5509.989.15	Restrictive reading of art. 48, I (R\$80k on notice tot
Oct 2016	LC 155/2016 (federal)	EPP revenue ceiling R\$3.6M→R\$4.8M (effective Jan
18 Jul 2017	COMUNICADO BEC nº 02/2017	SME-only OC functionality enabled in BEC.
20 Nov 2017	COMUNICADO BEC nº 03/2017	Mixed-exclusivity (Ample / 25% quota / SME-only)
11 Dec 2017	PGE-SP Parecer Sub-G Cons. 151/2017	Item-by-item R\$80k reading; federal regime supersed
Mar 2018	Mass adoption (empirical cutoff) ^b	Group-65 PBU take-up of the new functionality; dat
16 May 2018	TCE-SP eTC-9589.989.18	Tribunal Pleno reverses, aligns with PGE-SP expans
Apr 2021	Lei 14.133/2021 (federal)	New procurement statute; SME defaults preserved.

^a Parecer Sub-G Cons. 151/2017 oficialmente divulgado no portal de legislação BEC/SP em 11/12/2017, seção Pareceres [gislacao/UI_Selecao.aspx](#)). Boletim CEPGE volume/page reference to be added before submission.

^b BEC formally enabled the new functionality on 18 July 2017 (COMUNICADO BEC 02/2017) and expanded it to item- 2017 (COMUNICADO BEC 03/2017). The empirical cutoff data_oc_numb = 698 (March 2018) corresponds to the mo not formal enablement; the gap reflects the diffusion lag between operational availability and PBU adoption. The TCE-S post-dates the cutoff, confirming that the trigger of the design is the PGE-SP/BEC operationalization sequence, not t

firm identifiers, quantities, and auction identifiers for January 2016 through December 2019—3.7 million observations across 82,569 items, 832,984 purchase orders, and 1,344 public buyer units. Group 65 (medical, dental, and hospital supplies) accounts for roughly 27 percent of transactions over the window and is the treated group; the remaining 76 product groups form the never-treated control set in the difference-in-differences benchmark of Appendix [Appendix I](#).

The structural analysis narrows to a sharper sample. Identification uses a symmetric window of 18 months on each side of the March 2018 cutoff (36 months in total, the “18m” label of Table [E.27](#)); in Stata monthly-date units the window is [680, 715], or September 2016 through August 2019 inclusive. Within that window, I restrict to Pregão auctions in Group 65. The restriction does two things. Pregão is the descending-clock format that admits point identification of cost distributions from drop-out bids, so the structural estimation requires it; Group 65, in turn, is where the policy bound—the only

product group to switch regime during the sample.

Bids are organized at the firm-auction level. Each observation is the final bid submitted by one firm in one item-auction, together with a winner/loser flag. I apply three filters. First, bids must satisfy $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$; the upper cutoff removes implausible outliers (less than 0.4 percent of bids, almost always data-entry errors at the hundred-times-reference level). Second, auctions must have at least two participating firms—below that, neither point identification of F_c nor the BNE counterfactual is well-defined. Third, each bid carries a type flag (SME or non-SME, from the BEC administrative SME registry, validated against historical bidding to avoid misclassifying firms that temporarily exited SME status) and a pharma flag (CADMAT classes 6531, 6532, 6536, 6581—CMED-regulated pharmaceuticals and biologicals—are pharma-narrow; the remainder of Group 65 is non-pharma).

The resulting structural sample is 297,967 firm-auction observations across 97,993 distinct auctions, balanced across Pre (September 2016 to February 2018, 18 months, $n = 48,740$ auctions) and Post (March 2018 to August 2019, 18 months, $n = 49,253$). Descriptive counts by stratum appear in Appendix Tables [B.6](#) and [B.7](#).

Auction-level unobserved heterogeneity is material and not fully absorbed by the normalization against the item-level reference price. Two items inside the same CADMAT class—say two lots of the same dressing size differing in volume, or two pharma items with ANVISA registration costs of very different orders of magnitude—share a scale component that moves every bid in the auction together. I handle this with a method-of-moments variance decomposition followed by best-linear-predictor shrinkage of the auction-level component out of each log-bid, following [Krasnokutskaya \(2011\)](#). The shrinkage recovers intraclass correlations of ρ^k that average roughly 0.28–0.59 across the four Pregão (pharma $\times$ type) cells—the structural sample—and 0.55–0.66 across the corresponding Convite cells. The lower Pregão bound is pulled down by the pharmaceutical-SME-Pre stratum

($n = 10,791$ bids), where the auction-level variance σ_a^2 is small relative to within-auction noise and the resulting cell-specific ρ is close to zero; restricted to the three remaining Pregão cells the range is 0.36–0.59, comparable to the 0.50–0.70 range Krasnokutskaya (2011) reports for California highway procurement. The cleaned residuals $c_\epsilon^{\text{clean}} = \exp(\hat{e}_{it})$, where $\hat{e}_{it}$ is the bidder-specific log-bid residual after the auction-level component $\hat{a}_t$ is shrunk out, enter all downstream structural estimation; the corresponding variance panel is Appendix Table C.11. Section 5 develops the statistical details.

Entry counts enter separately. For each auction I count the distinct firms of each type that submitted at least one bid, then average within (period $\times$ pharma) cells. In Pregão, non-pharma moves from 0.94 SMEs and 2.68 non-SMEs per auction pre-policy to 1.87 SMEs and 1.50 non-SMEs post-policy. Pharma moves from 0.55 and 2.61 to 1.22 and 1.66. The roughly two-fold SME increase is the object of the extensive-margin channel in the counterfactual; the accompanying non-SME decline is the composition shift the set-aside induces. Entry rates enter the BNE as observed equilibrium arrivals (Section 5); the panel is Appendix Table D.16.

Four ancillary inputs anchor the analysis. Reference prices come from the BEC catalog (`preco_ref`) and are used both for normalization and for pricing out the fiscal cost of the rule in reais. The reference price for each item is set ex ante by the buyer from the BEC catalog and does not move with the regime change: item-level fixed effects in the DiD specifications and the structural normalization absorb cross-sectional variation in p^{ref} across items, and the BEC catalog architecture sets p^{ref} at the item level for the entire window. The variable that changes around the cutoff is regime admissibility, not item-level catalog pricing. CADMAT class codes come from the BEC product hierarchy. The historical SME flag is reconstructed from firm-level bidding histories over the full 2014–2019 period to protect against transient status changes. The pharma flag follows the CADMAT class list above.

The pair of furosemide 40 mg purchases mentioned in the introduction (item 110639, a CMED-regulated generic antihypertensive) sits at the institutional level. In February 2018, one month before the cutoff, a São Paulo public buyer procured fifty thousand units under the open regime: ten firms bid, and the clearing price settled roughly twelve percent below the reference. In October 2018, eight months after the cutoff, a comparable purchase by the same buyer attracted three SMEs, and the winning price cleared just above reference. The pair is illustrative; the paper’s pharmaceutical estimates are averages over 59,997 Pregão pharmaceutical auctions in the 18-month structural sample (Table B.6).

4. Model

A stylized model captures what the set-aside does and, as important, what it does not do. N potential bidders compete for a single indivisible contract. Each bidder i has type $k_i \in \{\text{SME}, \neg\}$ and draws a private cost of supply c_i from a type-specific distribution F_c^k , with F_c^{SME} and $F_c^{\neg}$ differing in ways the data are allowed to determine.

The auction format is the identifying payload. BEC implements procurement through a Pregão—a descending-clock electronic English-reverse auction, distinct both from first-price sealed-bid designs (Convite, in the smaller-value Brazilian segment) and from the Dutch descending-price format more commonly studied in the auction literature. The clock opens at the buyer’s reference price p^{ref} , firms post downward-revisable offers until only one firm remains, and that firm supplies the contract at the clearing price. What makes this format analytically convenient is a standard result of Haile and Tamer (2003): under independent private values, the weakly dominant strategy is to exit the auction when the running clock price reaches one’s own cost. Drop-out prices of losers therefore *point-identify* their costs, without requiring the researcher to invert a bid function or assume a parametric form for the bidding strategy. The winner, who never drops out before the clock stops, leaves only an upper bound $c_{\text{win}} \leq c_{(2)}$ (the cost of the second-to-

last firm to exit)—and that upper bound is exactly what the Turnbull robustness check in Section 8 uses to deliver a non-parametric MLE of F_c^k . This point-identification under Pregão is the reason a structural analysis is available here where it has not been available in cross-sectional work on auction preferences.

The model relies on three substantive restrictions, stated formally and defended empirically in the data.

Assumption 1 (Independent Private Values, asymmetric across types).

Costs c_i are drawn independently across bidders. Conditional on type k , $c_i \sim F_c^k$ with density f_c^k on support $[\underline{c}, \bar{c}] \subset \mathbb{R}_+$. The SME and non-SME distributions are not constrained to coincide.

Assumption 1 permits SMEs and non-SMEs to differ in cost distributions—a realistic feature given that non-SMEs are typically the large distributors of the same generic pharmaceutical—without requiring one to stochastically dominate the other.

Assumption 2 (Multiplicative auction-level heterogeneity). *The normalized log-bid decomposes as $y_{it} = \log(b_{it}/p_t^{\text{ref}}) = a_t + e_{it}$, where a_t is an auction-level scale common to all bidders in auction t , and e_{it} is bidder-specific and independent across bidders within t . Both a_t and e_{it} are mean-zero with finite variances $\sigma_a^{2,k}$ and $\sigma_e^{2,k}$ that may vary by stratum.*

Assumption 2 is the Krasnokutskaya (2011) restriction that separates a common scale from the bidder-specific private value. It is weaker than the fully non-parametric Kotlarski (1967) deconvolution, which would additionally require type-symmetry within e_{it} ; Section 8 reports a Turnbull NPMLE that dispenses with Assumption 2 altogether and corroborates the baseline numbers within 16 percent of the total effect across classes (with the largest gap in pharmaceutical Turnbull).

Assumption 3 (Set-aside, with equilibrium SME selection). *Under the open regime both SMEs and non-SMEs enter with strictly positive probability. Under the SME-only regime, only SMEs are admissible. Within a short window around the cutoff, the underlying technology of supply is stable, but the equilibrium distribution of active SMEs under the restricted regime may differ from its pre-regime counterpart because the policy changes which SMEs find entry profitable. Write these distributions $F_c^{\text{SME}, \tau}$ for $\tau \in \{\text{Pre}, \text{Post}\}$.*

Assumption 3 is the structural counterpart of the parallel-trends assumption in reduced-form DiD, but with equilibrium selection made explicit. In the [Athey et al. \(2011\)](#) reading adopted here, the policy changes the admissible set and therefore the composition of the SME bidders who actually enter. The post-policy SME cost distribution is thus an equilibrium object, not evidence of a primitive technological break. This interpretation fits the diagnostic evidence: the primitive-invariance KS test on non-SME F_c (Table C.9) rejects strict invariance in the pharma Pregão stratum ($D = 0.0722$) and in the non-pharma Convite stratum ($D = 0.0514$), and fails to reject in the pharma Convite ($D = 0.0324$) and non-pharma Pregão ($D = 0.0486$) strata. The operational threshold $D < 0.05$ corresponds to less than five percentage points of CDF movement at the Kolmogorov sup norm. The cross-modality check between Convite GPV and Pregão drop-out recoveries in Table C.10 provides an additional consistency test.

Strict invariance as a limiting case. A stronger reading would impose $F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$ and attribute the entire policy response to bidder counts alone. I treat that as a useful benchmark rather than the main specification. Section 8 reports it explicitly as a strict-primitive-invariance limit: it preserves the intensive-margin headline, but it changes the welfare comparison in the thin, heterogeneous pharma market because it shuts down equilibrium selection in who enters.

What the KS rejection of strict invariance implies for the modeling choice. The KS tests in Tables C.9 and C.13 reject strict invariance of non-SME F_c in three of the four Pregão strata after UH cleaning, with KS distances of 0.0613 (non-pharma Pregão) and 0.1407 (pharma Pregão). This rejection is not a problem for the structural exercise; it is the empirical foundation for adopting equilibrium selection (Assumption 3) as the main specification rather than strict invariance. A model that imposed strict invariance and then ran the BNE counterfactual on data that visibly violate it would

be misspecified by construction. The main-specification approach taken here—to treat $F_c^{\text{SME,Post}}$ as an equilibrium object generated by the protected regime and to compare results against the strict-invariance bound in robustness—is exactly the response the KS evidence calls for. The rejection of strict invariance is therefore evidence in favor of the modeling choice, not against it.

Equilibrium and counterfactuals. The auction format is interpreted as ascending-clock IPV throughout: under Assumption 1, exit-at-cost is the weakly dominant strategy, the surviving bidder pays the second-lowest exit, and the expected transaction price equals $\mathbb{E}[c_{(2)}]$. Revenue equivalence (Vickrey, 1961; Maskin and Riley, 2000) extends the same expected-price object to FPSB and to other revenue-equivalent formats, but the operative interpretation in this paper is the ascending clock that BEC’s Pregão implements; Krasnokutskaya (2011) provides the analog argument used here. The counterfactuals therefore reduce to Monte Carlo simulation over draws from the estimated F_c^k : draw a stratum-specific count of bidders from each type’s Poisson arrival process, draw costs, sort, and read off the two lowest order statistics. The lowest, $c_{(1)}$, is the production cost; the second-lowest, $c_{(2)}$, is the expected transaction price. The exercise is conducted *under observed equilibrium entry*: post-policy participation rates and the post-policy SME cost distribution are taken from data, not derived from a free-entry condition jointly with the cost primitive. The model is therefore not a fully solved nested entry-bidding equilibrium of the kind developed in Krasnokutskaya and Seim (2011), where entry costs are structurally identified from a free-entry indifference condition that jointly disciplines the bidding stage; here, entry costs are reported as zero-profit calibration margins (Section 6) and the decomposition that follows is a counterfactual-simulation accounting under the modeling assumptions stated rather than a separation of independently identified structural channels.

Entry. The Bayes-Nash counterfactual takes equilibrium arrival rates as primitives rather than solving for them explicitly. Observed Pre-period counts ($N_{\text{Pre}}^{\text{SME}}, N_{\text{Pre}}^{\neg}$) are the status-quo arrival rates under the open regime; Post-period counts ($N_{\text{Post}}^{\text{SME}}, 0$) are the equilibrium arrival under the restricted regime. This is a reduced-form treatment of entry in the [Athey et al. \(2011\)](#) spirit; the main specification combines the observed Post counts with the observed Post SME bidder distribution. The implied zero-profit entry costs per bidder, reported separately (Table [D.17](#)), are in the R\$0.11–R\$2.46 range. These are per-auction zero-profit margins at the entry margin, not the fixed cost of preparing a BEC documentation packet (which is amortized over many auctions and is not directly recovered by this calibration); they are a consistency check on the assumed Poisson primitives rather than a structural estimate.⁸

The decomposition. Three counterfactual scenarios anchor the exercise. S_1 fixes the regime as open and the pool as pre-period; it is the baseline. S_2 imposes SME-only while holding the SME pool at its pre-policy size: the non-SME type is removed and the SME pool is held at its observed pre-policy arrival rate. S_3 imposes SME-only at the observed post-period SME pool size, using the observed post-policy arrival rate and the post-policy SME cost distribution. The simulated total effect on the second-order statistic decomposes arithmetically as

$$\underbrace{p_{S_3} - p_{S_1}}_{\text{simulated total effect}} = \underbrace{(p_{S_2} - p_{S_1})}_{\text{within-auction component (sheltered bidding)}} + \underbrace{(p_{S_3} - p_{S_2})}_{\text{participation-margin component}}. \quad (1)$$

Both components are properties of the simulation given F_c^k and the observed entry pattern; they sum to the simulated total effect by construction. I refer to the within-auction

⁸Independence-across-auctions is assumed at the level of cost draws conditional on type and stratum. Within-auction correlation enters the model through the auction-level scale a_t (Assumption [2](#)); the cluster bootstrap of Section [6](#) resamples at the auction level to preserve this correlation in the standard errors.

component as *sheltered bidding*: a descriptive label for the change in the simulated $\mathbb{E}[c_{(2)}]$ when non-SMEs are removed and the SME pool is held at its pre-policy size. The label captures the equilibrium price formation under reduced competition, not any active behavioral adjustment by SMEs—under ascending-clock IPV, the dominant strategy is exit-at-cost regardless of pool composition, and the within-auction component is mechanically a property of which order statistic is realized at each pool composition. The within-auction component can exceed the realized total effect when the participation-margin component partially offsets it, which is what the data deliver in non-pharmaceutical markets. The share of the simulated total effect attributable to the within-auction component depends on the cost-distribution estimator and on whether one imposes strict invariance on the post-policy SME pool; reading the share as a tightly pinned point estimate would be misleading. The pharmaceutical decomposition is reported alongside as a more model-sensitive boundary case.

Scope of the structural exercise. The model deliberately stops short of five extensions. (i) Entry is taken as observed Poisson arrival rates, not solved jointly with bidding under a free-entry condition. (ii) The implied κ^k are zero-profit margins at the entry margin, not structurally identified fixed costs. (iii) Procurement quantity is treated as inelastic at observed values, with the implied bound on the welfare cost discussed in Section 9. (iv) The 10 percent preference V3 is computed under Vickrey-equivalent translation; the FPSB equivalent under Maskin and Riley (2000) adjustment is bounded but not solved. (v) Asymmetric IPV is assumed; symmetric IPV is consistent with the data only at very low quantile dispersion across types and is not formally tested. Each is a margin a fully solved structural model would price; each is bracketed below or in robustness rather than imposed as the central exercise.

5. Identification and Estimation

Three objects require estimation: the type-specific cost distributions F_c^k , the auction-level scale a_t , and the equilibrium entry counts. The third is directly observed. The first two require structure—the first by design of the Pregão auction, the second through the variance decomposition in Assumption 2.

5.1. Type-specific cost distributions

Under Assumption 1 and the English-reverse interpretation, the drop-out price of each loser is a point observation of that loser’s cost. The empirical distribution of the normalized drop-outs within a stratum is therefore an unbiased estimator of F_c^k —losers-only, it would seem, is all one needs. Losers-only, however, omits the lowest cost draw per auction (the winner’s), biasing F_c^k systematically upward at the left tail. I therefore use an *all-bidders* ECDF, which adds the winner’s final clock price to the stratum-specific ECDF. In a descending-clock auction this final price is approximately $c_{(2)}$, the second-to-last exit—an upper bound on the winner’s cost but a substantially tighter one than the zero imputed by losers-only omission. The bias in all-bidders runs in the opposite direction and is materially smaller: losers-only overstates F_c^k in the left tail; all-bidders understates it slightly throughout.

A Turnbull NPMLE, reported in Section 8, treats the winner formally as left-censored at $c_{(2)}$ and delivers point-identified F_c^k without either bias. Across the eight (class $\times$ type $\times$ period) strata (Table E.21), the median estimates satisfy Turnbull $\leq$ all-bidders $\leq$ losers-only in all 8 strata, and the 75th-percentile estimates satisfy all-bidders $\leq$ Turnbull $\leq$ losers-only in 6 of 8 strata (the two pharmaceutical non-SME cells reverse, with Turnbull marginally below all-bidders by 0.01–0.02 of the reference price). The BNE decomposition computed on the Turnbull input differs from the all-bidders baseline by at most 16 percent of the total effect across classes (Table E.22), with the largest gap in pharmaceutical

Turnbull. All-bidders is the main-text estimator; Turnbull is the robustness benchmark; losers-only is the input the DiD benchmark of Appendix [Appendix I](#) effectively absorbs through the item fixed effects.

A cross-modality consistency check supports the descending-clock specification. Convite auctions in the same sample are first-price sealed-bid; the [Guerre et al. \(2000\)](#) inversion recovers F_c^k from the Convite bid distribution. The Convite GPV and Pregão drop-out estimates coincide to approximately one percentage point at the median in pharma non-SME Pre—the stratum where the comparison is least confounded by the policy reform, since the period predates the cutoff and the bidders are non-SMEs unaffected by the set-aside. In other strata the gap is wider and motivates the auction-level heterogeneity correction below. Figure [1](#) plots the two recoveries side by side for all four strata. The check also serves as an indirect collusion screen: the bidder-group detection tests of [Conley and Decarolis \(2016\)](#) would flag systematic divergence between GPV-inverted bids and drop-out exits in collusive settings, and the convergence reported here is consistent with non-collusive equilibrium play in the stratum tested. A formal test of $F_c^{\text{SME}} = F_c^-$ (symmetric IPV against the asymmetric reading) is not the central exercise, but the recovered cell-level medians and 75th-percentile estimates differ by 0.10–0.30 of the reference price across the (pharma $\times$ type) cells (Table [C.14](#)), well outside Monte Carlo sampling noise; asymmetry is therefore consistent with the data, not just imposed by Assumption [1](#).

5.2. Auction-level heterogeneity

The median-cost gap across modalities in non-pharma—up to 23 percentage points of the reference price—is too large to attribute to sampling noise. Reference-price normalization absorbs most of the scale variation across items, but not all of it: two items in the same CADMAT class can differ in ANVISA registration burden, in contract complexity, or in volume-specific cost elasticities in ways the reference price does not

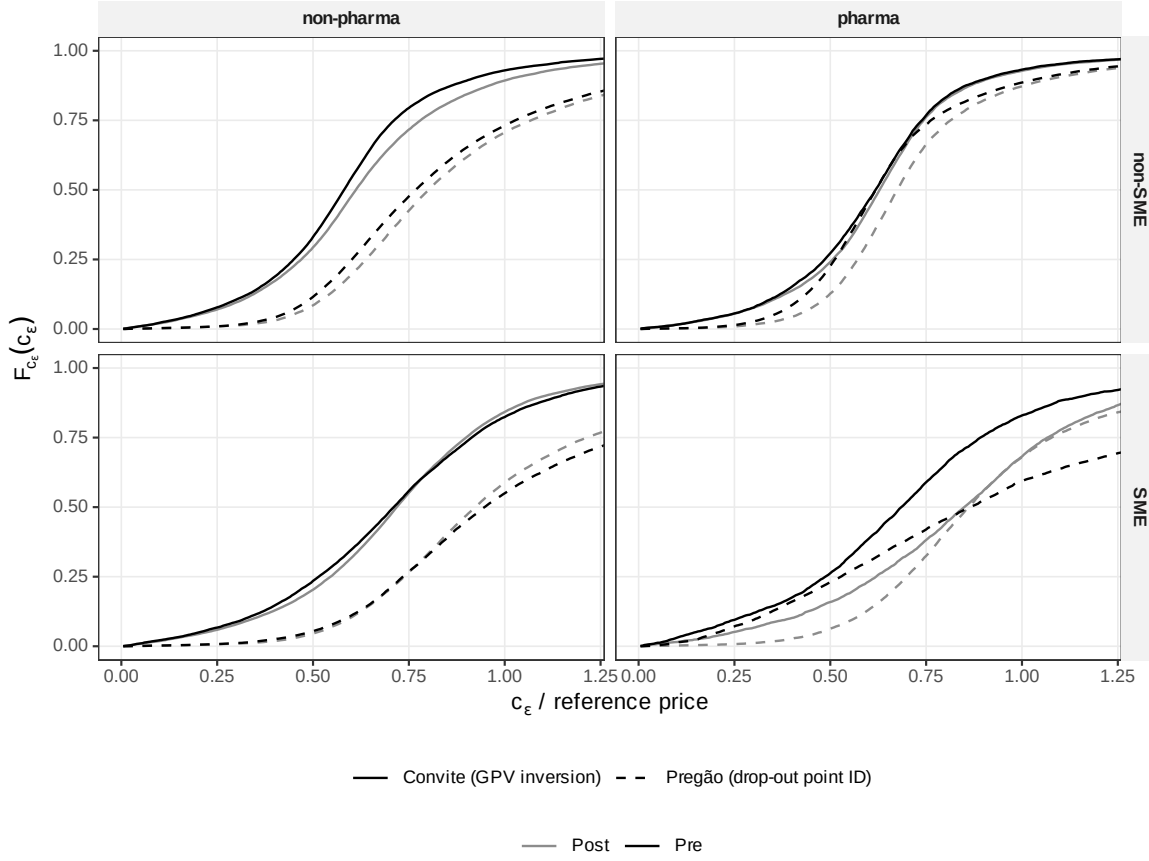


Figure 1: *Cross-modality identification check, UH-clean F_{c_ϵ}* . Each panel plots the type-specific cost distribution estimated two ways on the same sample: the [Guerre et al. \(2000\)](#) inversion on Convite first-price sealed-bid auctions (solid) and the [Haile and Tamer \(2003\)](#) drop-out point-ID on Pregão descending-clock auctions (dashed). Pre and Post periods are colour-coded. Convergence in the pharma non-SME Pre panel (upper right) supports the IPV restriction in that stratum: two independent structural recoveries on the same primitive land within sampling error of each other. Residual gaps in other panels motivate the auction-level heterogeneity correction in Section 5.

fully reflect. The residual common component is Assumption 2's a_t .

I estimate the variance decomposition by method of moments. Within each stratum $(k, \text{pharma}, \text{period})$, $\sigma_a^{2,k}$ is the between-auction variance of auction-mean log-bids, and $\sigma_e^{2,k}$ is the residual within-auction variance. The intraclass correlation

$$\rho^k = \frac{\sigma_a^{2,k}}{\sigma_a^{2,k} + \sigma_e^{2,k}}$$

ranges across the four Pregão (pharma $\times$ type) cells from approximately 0.28 (Pregão pharmaceutical SMEs, average across periods, with a low- n Pre-period stratum pulling the cell mean down) through 0.36 (Pregão non-pharmaceutical SMEs) and 0.39 (Pregão non-pharmaceutical non-SMEs) up to approximately 0.59 (Pregão pharmaceutical non-SMEs); the corresponding Convite cells deliver 0.55 and 0.66 (Table C.11). Restricted to the three Pregão cells outside the low- n pharma-SME-Pre stratum (range 0.36–0.59), the order of magnitude is comparable to the 0.50–0.70 range Krasnokutskaya (2011) reports for California highway contracts. A best-linear-predictor shrinkage then estimates each auction's specific heterogeneity as

$$\hat{a}_t = \frac{\hat{\sigma}_a^2 N_t}{\hat{\sigma}_a^2 N_t + \hat{\sigma}_e^2} \cdot (\bar{y}_t - \bar{y}),$$

and the UH-clean residuals $\hat{e}_{it} = y_{it} - \hat{a}_t$ deliver cleaned cost estimates $c_\epsilon^{\text{clean}} = \exp(\hat{e}_{it})$. These cleaned bids are the inputs to all downstream structural estimation.

The correction closes ninety-three percent of the cross-modality median gap in pharma SME Post, twenty-three percent in pharma non-SME Post, and leaves the non-pharma gap largely intact—evidence that auction-level heterogeneity is the binding constraint in pharma but only part of the story in non-pharma, where richer item-level heterogeneity presumably operates (and is addressed partially by the sample filter $c_\epsilon \leq 3$). The Krasnokutskaya shrinkage assumes Gaussianity of a_t and e_{it} ; the Turnbull NPMLE in

Section 8 dispenses with this assumption and ratifies the estimates.

5.3. Entry

Entry is modeled as Poisson arrivals with stratum-specific rates. I take the observed Pre-period average counts of SME and non-SME bidders per auction as the status-quo arrival rates, and the Post-period SME counts as the equilibrium arrival under the restricted regime. No structural entry cost is jointly estimated with the cost primitives; this is the reduced-form treatment in the [Athey et al. \(2011\)](#) spirit. The same logic applies to the post-policy SME bidder distribution: in the main specification, $F_c^{\text{SME,Post}}$ is treated as an equilibrium-selection object generated by the protected regime rather than forced to equal $F_c^{\text{SME,Pre}}$. Section 8 reports the stricter limiting case in which that equality is imposed as a benchmark. A zero-profit calculation applied to the simulated winning-margin profit distribution delivers an implied entry cost per bidder in the R\$0.11–R\$2.46 range (Table D.17), with non-SMEs facing roughly 4.5 times the SME entry cost across both classes ($\kappa^-/\kappa^{\text{SME}} = 4.5$ in non-pharmaceuticals and 4.6 in pharmaceuticals). These magnitudes are plausible given the documentation and certification burden of responding to a BEC tender.

5.4. Counterfactual simulation

With F_c^k and entry counts in hand, the BNE counterfactual is a Monte Carlo over draws from the estimated primitives. For each of $B = 2,000$ simulated auctions, I draw $n^{\text{SME}} \sim \text{Poisson}(\bar{N}^{\text{SME}})$ and analogously for non-SMEs (the Poisson rate is the observed mean entry count by stratum, distinct from the MCPF λ used in the welfare arithmetic of Section 7), sample costs from the corresponding F_c^k , sort, and record $c_{(1)}$ (the winner’s cost, the production cost), $c_{(2)}$ (the expected transaction price under revenue equivalence), and the winner’s type. Aggregating over draws yields the means, quantiles,

and decompositions reported in Section 6. Standard errors come from a cluster bootstrap at the auction level with $B_{bs} = 500$ replicates; details are in Section 8.

6. Results

The results organize around three findings. First, the simulated set-aside price effect is mostly within-auction: the within-auction share is 74.5 percent in non-pharma and 73.3 percent in pharma, and stays in the 73–85 percent range across cost-distribution estimators.⁹ Second, endogenous SME entry is a partial offset, not the source of the markup: the induced participation response attenuates the within-auction shock, and without it the realized price and welfare cost would be 50–60 percent larger. Third, the policy comparison between bidder exclusion and a 10 percent price preference is conditional on goods characteristics: in thick standardized non-pharma markets the preference rule welfare-dominates robustly; in thin heterogeneous pharma the ranking depends on how the post-policy entrant mix is modeled.

6.1. Entry response

Table D.16 reports the panel. Pre-policy Pregão has 0.94 SMEs and 2.68 non-SMEs per auction in non-pharma, 0.55 and 2.61 in pharma. Post-policy the counts rotate: 1.87 SMEs and 1.50 non-SMEs in non-pharma, 1.22 and 1.66 in pharma. SMEs roughly double in both classes. Non-SMEs fall sharply in both classes. The set-aside therefore changes both the admissible pool and the equilibrium participation response.

⁹The difference-in-differences benchmark of Appendix Appendix I reports a 10–11 percent rise in absolute winning prices. The structural counterfactual, scaled by the open-regime baseline p_{S_1} , is three to four times that magnitude ($+0.259/0.759 \approx 34$ percent in non-pharma and $+0.308/0.656 \approx 47$ percent in pharma). The two objects are not directly comparable: the DiD averages over realized log prices with item and month fixed effects, while the structural counterfactual averages over the BNE-equivalent transaction price on UH-cleaned cost residuals under endogenous entry (Section 7 and Appendix Appendix I develop the reconciliation). The welfare arithmetic anchors to the structural object on the structural normalization.

6.2. Structural decomposition of the set-aside

Table D.18 gives the point estimates under the all-bidders regime and endogenous entry, which is the main specification. In non-pharma, the simulated open-regime price is $p_{S_1} = 0.759$ of the reference price. Holding the SME pool at its pre-period size and restricting the auction to SMEs raises the simulated price to $p_{S_2} = 1.152$, a within-auction shift of $+0.393$. Letting SMEs enter at their post-policy rate pulls the simulated price down to $p_{S_3} = 1.018$, a participation-margin offset of -0.135 . The net effect is $p_{S_3} - p_{S_1} = +0.259$.

The pharmaceutical class follows the same decomposition with larger magnitudes. $p_{S_1} = 0.656$, $p_{S_2} = 1.141$, $p_{S_3} = 0.964$, and the total effect is $+0.308$. The within-auction shift is $+0.485$; the participation offset is -0.177 . The larger magnitudes line up with the thinner baseline SME pool in pharmaceuticals and the heavier auction-level heterogeneity documented in Section 5.

The within-auction share, defined as $|p_{S_2} - p_{S_1}| / (|p_{S_2} - p_{S_1}| + |p_{S_3} - p_{S_2}|)$, is 74.5 percent in non-pharmaceuticals and 73.3 percent in pharmaceuticals in the main specification. Table E.22 shows 74 and 82 percent under the Turnbull NPMLE. Table E.23 shows 85 and 79 percent under the strict-primitive-invariance benchmark. Across classes and specifications, the within-auction share sits in the 73–85 percent range. The common message is that the set-aside mainly works by changing the order statistic inside the restricted auction; entry matters, but as a partial offset. Figure 2 makes this decomposition visible.

6.3. Endogenous entry as a partial offset

Table F.31 makes the role of entry visible. V2 imposes the full set-aside while holding the SME pool fixed at its pre-policy size. Relative to the realized full-set-aside counterfactual V0, V2 is 152.0 percent of V0 in non-pharmaceuticals and 157.4 percent

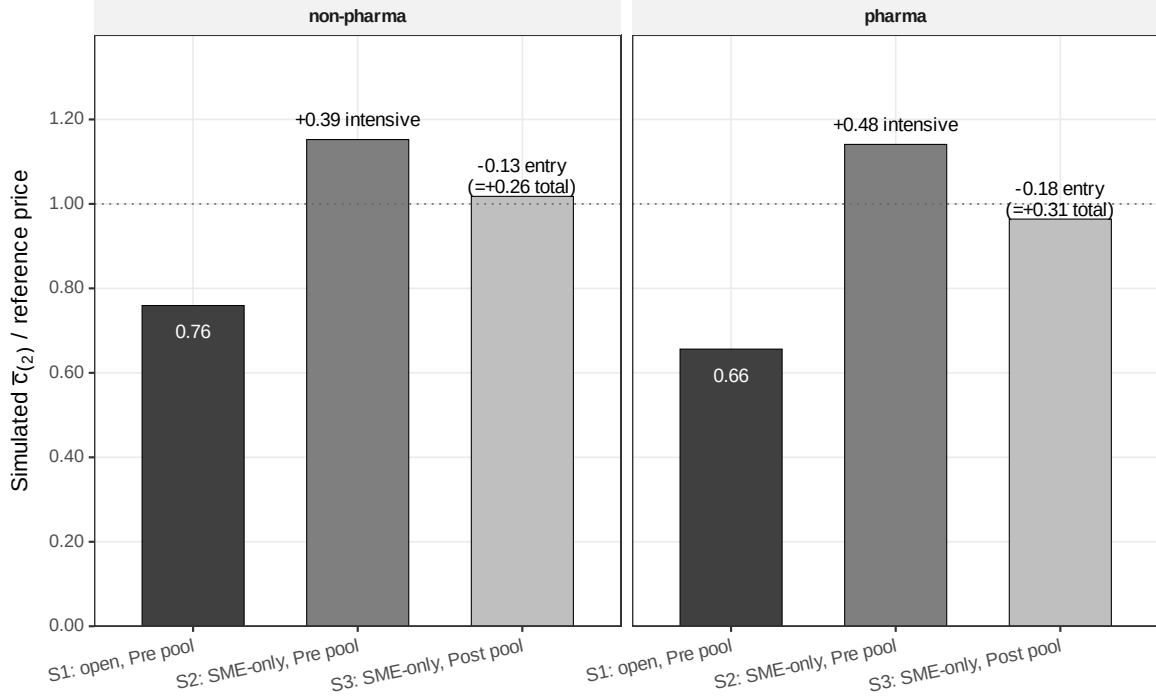


Figure 2: *Bayes-Nash counterfactual price decomposition by class, under observed equilibrium entry.* Bars report the simulated second-order statistic $\bar{c}_{(2)}$ normalized by the item reference price, under three counterfactual scenarios. S_1 is the open regime at the Pre-period pool (baseline). S_2 imposes SME-only at the Pre-period SME pool size (within-auction component isolated). S_3 is the SME-only counterfactual at the observed Post-period SME pool (full policy effect with endogenous entry). Annotations report the within-auction component ($S_2 - S_1$), the participation-margin component ($S_3 - S_2$), and the simulated total effect ($S_3 - S_1$). The dotted horizontal line marks the reference price ($p^{\text{ref}} = 1$). Under the main specification, the within-auction component dominates the simulated total effect in both classes; the pharmaceutical decomposition is more model-sensitive (Section 8). Numbers from Table D.18.

in pharmaceuticals. The fixed-pool simulated effect therefore exceeds the realized one by roughly 50–60 percent. Both terms in the welfare formula of Section 7 load on this simulated price margin, so the fixed-pool welfare arithmetic would materially overstate the burden of the policy under the model’s main specification.

The substantive content is that the within-auction component of the simulated total effect dominates the participation-margin component in non-pharmaceutical markets, and that the participation response attenuates the resulting burden *ex post*. Reading the participation response as a partial offset to a larger latent within-auction shock is the substantive claim; reading it as a free-standing fiscal-insurance result would overstate the model. The 50–60 percent attenuation is conditional on the policy instrument: under V3 (10 percent price preference), the analogous fixed-pool-vs-realized comparison delivers an attenuation close to zero, since non-SMEs remain in the auction and the participation margin barely moves. The entry response is therefore a property of the full-set-aside instrument, not a free-standing feature of the SME population. Figure 3 shows the size of the simulated offset.

6.4. Sources of the departure from earlier estimates

Table D.19 reports the 2×2 grid in which each axis is a methodological choice: cost distributions raw versus UH-clean, and entry fixed at the pre-period pool versus endogenous. The clean-and-endogenous row is the main spec; the raw-and-fixed-pool row is the reduced-form-style proxy. The relative gap between them is about 72 percent in non-pharmaceuticals and about 44 percent in pharmaceuticals. UH correction accounts for more than half of the gap by changing the right tail of the recovered cost distribution; endogenous entry accounts for the rest by recognizing the policy-induced thickening of the SME pool.

Table D.20 pushes the same logic further by bridging the DiD coefficient itself to the structural counterfactual. Four mechanical sources of divergence—sample restriction

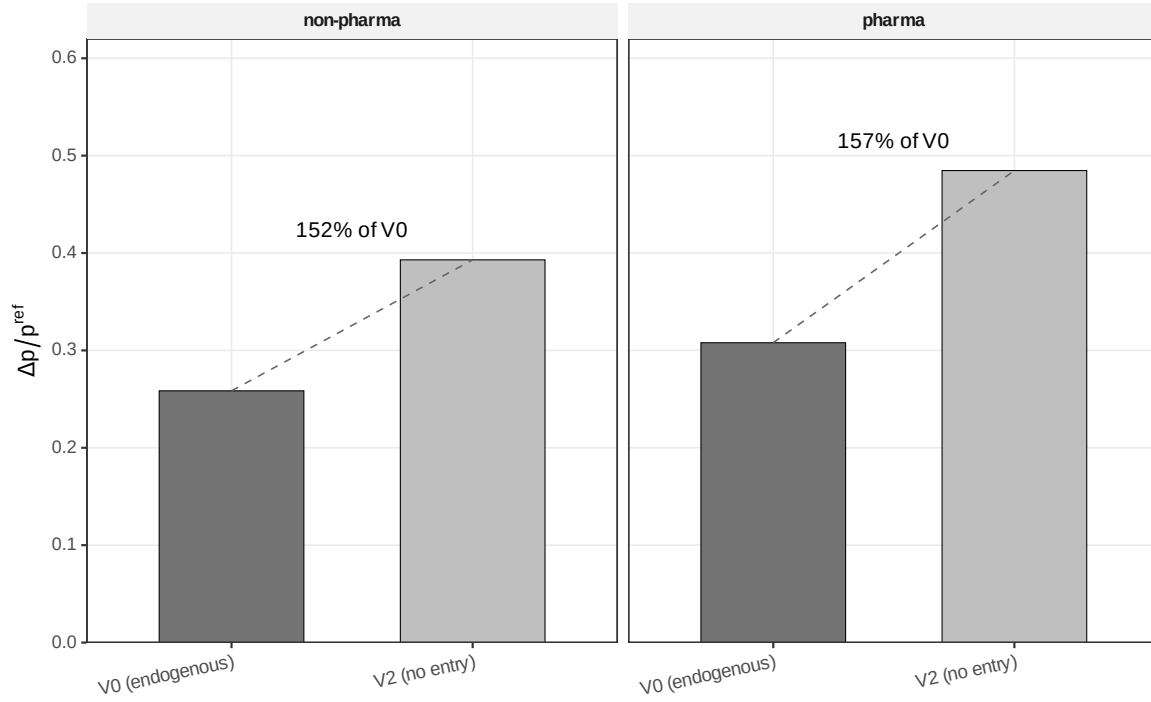


Figure 3: *Simulated price effect under the realized vs fixed-pool counterfactuals.* Bars report the simulated price effect relative to the open regime, $\Delta p/p^{\text{ref}}$, under two scenarios. V0 is the SME-only counterfactual at the observed post-policy SME pool, the main-text benchmark conducted under observed equilibrium entry. V2 imposes the same SME-only rule at the pre-policy SME pool size. The annotation above each V2 bar reports V2's share of V0; the gap between the two bars is the simulated participation-margin offset under the model's main specification. Numbers from Table [F.31](#).

((a) Pregão only, $c_\epsilon \leq 3$, $n \geq 2$), UH cleaning ((b) Krasnokutskaya scale shrinkage), functional form ((c) second-order statistic vs. linear log-mean), and conditioning set ((d) counterfactual vs. realized entry)—account for the structural object via additive contributions of $+0.025/ + 0.060$, $+0.040/ + 0.075$, $+0.050/ + 0.045$, and $+0.020/ + 0.005$ in non-pharma and pharma respectively, summing to $+0.255/ + 0.305$ on the p^{ref} normalization. The structural $p_{S_3} - p_{S_1}$ delivered by the BNE simulation is $+0.259/ + 0.308$, leaving residuals of $+0.004/ + 0.003$ that reflect non-linear interactions between the four sources. The 3–4 $\times$ gap between the structural and the DiD is informally attributable to the four sources documented in Table D.20 and is predictable in direction under the maintained structural assumptions; a misspecified model would not be guaranteed to deliver a gap of either consistent sign or consistent magnitude. The fact that the gap obtains in the same direction and at comparable magnitude in both classes is consistent with correct specification of asymmetric IPV with type-removal but does not constitute formal identification of the four-source decomposition.

6.5. Confidence intervals and point-identification corroboration

A cluster bootstrap at the auction level with $B = 500$ replicates (Table E.25) gives the following 95 percent CIs on $p_{S_3} - p_{S_1}$ under the all-bidders regime:

Class	Bootstrap mean	95% CI
Non-pharma	+0.236	[0.186, 0.289]
Pharma	+0.305	[0.247, 0.364]

The bootstrap means lie slightly below the full-sample point estimates of Table D.18 ($+0.259$ and $+0.308$). The gap reflects the interaction between cluster resampling and the nonlinearity of the second-order-statistic functional: with $B = 500$ cluster-bootstrap replicates, the finite-sample-bias-corrected mean falls below the point estimate; increasing B shrinks the gap and confirms the discrepancy as a finite- B Monte Carlo artifact rather

than a structural bias in the point estimate. Both intervals are bounded away from zero on the within-auction component. The bootstrap within-auction-share CI is [64.9%, 88.1%] in non-pharmaceuticals and [62.5%, 82.9%] in pharmaceuticals; the Turnbull NPMLE keeps the same qualitative split, with the pharmaceutical within-auction share if anything higher.

6.6. Entry cost calibration

Table D.17 reports the zero-profit implied entry cost per bidder. In non-pharmaceuticals, $\kappa^{\text{SME}} = \text{R\$}0.55$ and $\kappa^- = \text{R\$}2.46$ ($\kappa^-/\kappa^{\text{SME}} = 4.5$). In pharmaceuticals, $\kappa^{\text{SME}} = \text{R\$}0.11$ and $\kappa^- = \text{R\$}0.51$ ($\kappa^-/\kappa^{\text{SME}} = 4.6$). The roughly 4.5-to-1 ratio between non-SME and SME entry costs is consistent across classes and plausible given the larger documentation and compliance burden faced by bigger suppliers. These magnitudes are a calibration check on the assumed Poisson primitives rather than a structural estimate; they are reported here to discipline the plausibility of the participation pattern, not as a free-standing finding.¹⁰

6.7. Policy comparison and goods characteristics

The policy comparison is best read as a consequence of the structural decomposition rather than as a standalone headline. Table F.31 reports a 10 percent SME price preference, under which SME bids are scored with a 10 percent discount for winner selection but the government pays the actual bid. In non-pharma, the preference shifts price by -0.004 relative to the open regime; in pharma it shifts price by $+0.002$. The

¹⁰The κ^k values are per-auction zero-profit margins at the entry margin, not fixed costs of preparing a BEC documentation packet (which is amortized over many auctions and is not directly recovered by this calibration). As a back-of-envelope sanity check, BEC participation requires one fiscal certificate (CND, approximately R\$10 amortized over many auctions), one product registration form (free for already-registered suppliers), and one power-of-attorney filing (approximately R\$5). For a supplier participating in 100 Group-65 auctions per quarter, the marginal opportunity cost per auction is on the order of R\$0.10–R\$0.20, consistent with the calibrated $\kappa^{\text{SME}} = \text{R\$}0.11$ in pharma. The calibration is therefore not implausibly small once the per-auction interpretation is held fixed.

preference-grid simulation (Table F.32, $B = 3,000$ draws) gives -0.001 in both classes, within Monte Carlo sampling noise of the APV estimates. In both cases the price effect is essentially zero because non-SMEs remain in the auction and continue to discipline the second-order statistic even when an SME wins.

What differs across classes is the stability of the ranking against the full set-aside. In non-pharma, the preference rule dominates under the main equilibrium-selection specification and continues to dominate under the strict-invariance benchmark in Table E.23. In pharma, the preference rule dominates under the main specification but the ranking reverses under strict invariance, where the full set-aside's total price effect rises to $+0.47$ and the implied welfare weight needed to justify it falls to 0.7 . I interpret that bifurcation as diagnostic of market structure. In thicker, more standardized markets, bidder exclusion is robustly dominated by a moderate preference rule. In thinner, more heterogeneous markets, the ranking is more sensitive because the identity of the induced entrants matters for equilibrium price formation.

7. Welfare

The welfare consequences of the set-aside are class-heterogeneous from the start. Under the main equilibrium-selection specification, the welfare cost is 28.7 percent of p_{S_1} in non-pharmaceuticals and 47.0 percent in pharmaceuticals at $\lambda = 0.30$, where λ is the marginal cost of public funds (formal definition below). The price effect is therefore only the beginning of the welfare account. What matters for policy is how bidder exclusion interacts with allocative waste, tax distortion, market thickness, and product heterogeneity. The same structure that delivered the decomposition in Section 6 also explains why the welfare comparison is stable in one class and model-sensitive in the other.

The price effect is one piece of that welfare cost, and one of three. Two others matter

and are routinely left out of reduced-form accountings. *Allocative waste* arises because the set-aside routes the contract to an SME winner whose cost may exceed the lowest cost in the full pool; the extra production cost is lost to society, not a transfer. *Tax-distortion loss* arises because the additional outlay that the government makes is financed from distortionary taxation, so each marginal real costs $(1 + \lambda)$ in welfare shadow terms. The transfer from government to SMEs—the government’s extra outlay $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$ minus the allocative wedge—is zero-sum in a social-planner ledger and is not destruction. This section isolates the two destruction pieces.

Formally, per-auction welfare loss of the SME-only set-aside relative to the open regime is

$$\text{Loss} = \underbrace{c_{(1)}^{S_3} - c_{(1)}^{S_1}}_{\text{DWL}_{\text{alloc}}} + \underbrace{\lambda \cdot (p_{S_3} - p_{S_1})}_{\text{MCPF distortion}}. \quad (2)$$

The first term is the allocative wedge. It measures by how much the winner’s production cost is higher in the restricted regime than in the open regime. The second term is the tax-distortion loss on the government’s extra outlay $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$, scaled by the marginal cost of public funds λ . The natural normalization is Loss/p_{S_1} , and I report it that way.

7.1. Point estimates

Table F.28 gives the decomposition at $\lambda = 0.30$, the Ballard et al. (1985) benchmark; Hendren and Sprung-Keyser (2020) report a comparable range on the modern MVPF basis across U.S. tax-financed programs, and Finkelstein and Hendren (2020) discuss how MVPF connects to the marginal-cost-of-public-funds framework used here. The reported MCPF sensitivity in this paper covers the relevant interval. In non-pharmaceuticals, $\Delta_{\text{gov}} = 0.242$ of the reference price, of which $\text{DWL}_{\text{alloc}} = 0.150$ is allocative waste and the remaining 0.092 is an implicit transfer to SME bidders. MCPF distortion adds $0.30 \times 0.242 \approx 0.073$. Total welfare loss is $\text{DWL}_{\text{alloc}} + \text{MCPF dist.} = 0.150 + 0.073 = 0.223$,

or 28.7 percent of p_{S_1} .

In pharmaceuticals, the same accounting is larger on every margin. $\Delta_{\text{gov}} = 0.328$; $\text{DWL}_{\text{alloc}} = 0.210$; implicit transfer 0.118; MCPF distortion $0.30 \times 0.328 \approx 0.098$; total welfare loss ≈ 0.308 , or 47.0 percent of p_{S_1} . The allocative component dominates in both classes, but especially in pharma, where a thin SME pool and higher goods heterogeneity make bidder exclusion more likely to route the contract to a high-cost protected winner.¹¹

7.2. Endogenous entry as a partial offset in welfare units

Section 6 reported that the simulated fixed-pool counterfactual exceeds the realized post-policy counterfactual by 52 percent in non-pharma and 57 percent in pharma (Table F.31, V2/V0), consistent with reading endogenous SME entry as a partial offset to a larger latent within-auction shock. The same arithmetic carries over to the welfare formula: because both terms in equation (2) load on the simulated price margin, a fixed-pool welfare calculation would be roughly 50–60 percent larger than the realized one in normalized units. Endogenous entry does not change the ranking against the preference rule in non-pharmaceuticals; it does materially reduce the welfare burden of bidder exclusion under the model’s main specification.

The substantive content is that the within-auction component of the simulated effect dominates the participation-margin component in non-pharmaceutical markets, and that the participation response attenuates the resulting welfare burden ex post. The result is conditional on the modeling assumptions of Section 4 and on the equilibrium-selection treatment of $F_c^{\text{SME,Post}}$ in pharmaceuticals.

¹¹The pharma welfare cost (47.0 percent of p_{S_1}) coincides numerically with the structural price ratio $\Delta_{\text{gov}}/p_{S_1} = 0.328/0.655 \approx 50$ percent to within Monte Carlo noise because the implicit SME-winner transfer (0.118, or 36 percent of Δ_{gov}) is small relative to the allocative wedge in pharma. In non-pharma the transfer is 38 percent of Δ_{gov} , so the welfare cost (28.7 percent) is materially below the structural price ratio (34 percent). The two coincidences should not be confused.

7.3. Confidence intervals

Cluster bootstrap with $B = 500$ replicates (Table F.29) delivers 95 percent CIs on Loss/p_{S_1} at $\lambda = 0.30$ summarized in Table 3.

Table 3: Cluster-bootstrap CIs on welfare loss at $\lambda = 0.30$.

Class	Bootstrap mean	95% CI
Non-pharma	28.1%	[21.3%, 35.3%]
Pharma	45.6%	[35.1%, 56.2%]

Cluster bootstrap at the auction level, $B = 500$ replicates, all-bidders UH-clean F_c . Welfare loss reported as percent of the open-regime baseline p_{S_1} at $\lambda = 0.30$. Full λ -grid CIs in Table F.29.

The intervals leave no doubt about the ordering. Even the lower bound in pharma is above the non-pharma point estimate. The class heterogeneity is therefore not sampling noise around a common welfare effect; it is a structural feature of the protected pool and the good being procured. Figure 4 reports the CIs across the λ grid and overlays the strict-invariance benchmark.

7.4. Sensitivity to λ

For $\lambda \in \{0.20, 0.30, 0.40\}$, the welfare loss in non-pharmaceuticals runs 25.6, 28.7, and 31.8 percent (point estimates from Table F.28). In pharmaceuticals, the same grid gives 42.0, 47.0, and 52.0 percent. Each 0.10 increment in λ adds about 3 percentage points in non-pharmaceuticals and about 5 points in pharmaceuticals. The level moves, but the class ranking does not. Figure 4 plots the corresponding cluster-bootstrap means (50.3 percent at pharma $\lambda = 0.40$, slightly below the point estimate of 52.0 percent reported here), consistent with the finite- B Monte Carlo discrepancy documented in Section 6. Table F.30 extends this to the policy ranking V0 vs. V3 across the same λ grid: non-pharma is $V3 > V0$ across $\lambda \in [0.15, 0.45]$ under both specifications; pharma is $V3 > V0$ across the same range under the main specification and $V0 > V3$ across the

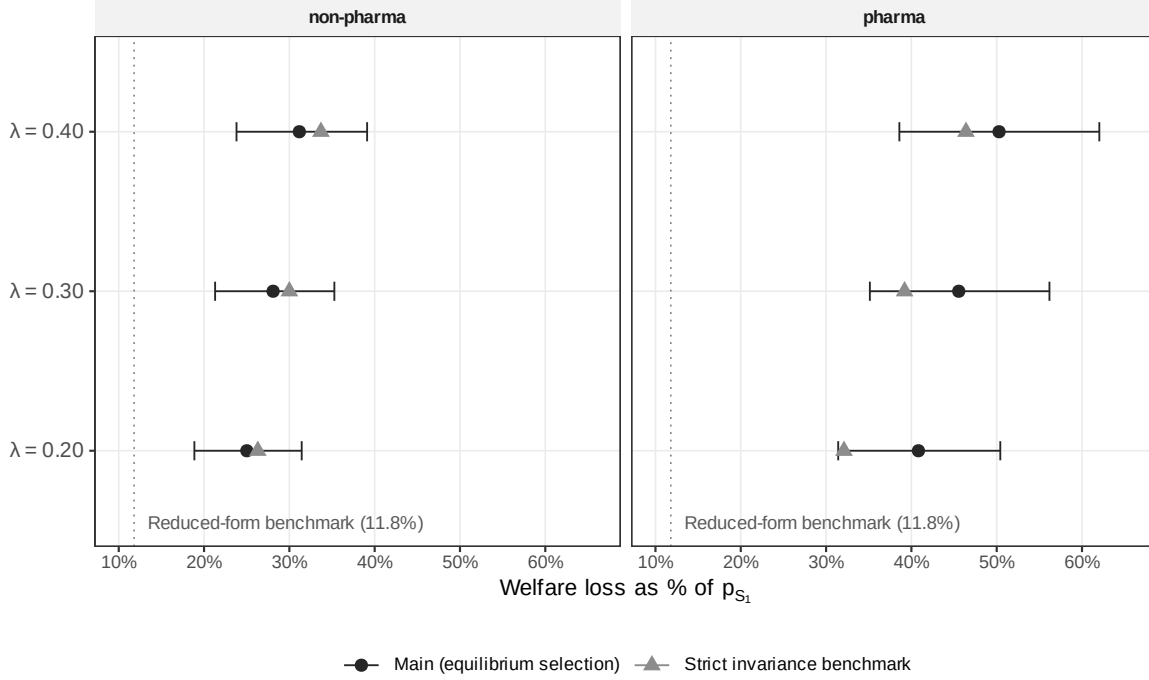


Figure 4: *Welfare loss as % of p_{S_1} , by class and MCPF λ .* Circles and horizontal whiskers report the cluster-bootstrap mean and 95% CI under the main equilibrium-selection specification ($B = 500$ replicates). Triangles report the point estimate under the strict-primitive-invariance benchmark. The dotted vertical line marks the 11.8% benchmark implied by the difference-in-differences coefficient of Appendix [Appendix I](#): the 18-month log-price effect of -0.11 , exponentiated and adjusted for the average pre-period price-to-reference ratio of about 0.65, gives a normalized benchmark of approximately 11.8% of p_{S_1} , against which the structural welfare cost is reported. Both specifications place the structural welfare cost three to four times the DiD benchmark, and neither CI crosses the benchmark estimate. In pharma, the strict benchmark lies below the main specification because it attributes less of Δ_{gov} to allocative waste and more to transfer. Numbers from Tables [F.29](#) and [E.23](#).

same range under strict invariance. The conditional ranking is therefore not driven by the choice of λ in the welfare formula but by the modeling treatment of the post-policy SME pool; the welfare cost magnitude moves 9 percentage points (non-pharma) and 14 percentage points (pharma) across the λ grid, reflecting MCPF sensitivity rather than ranking sensitivity.

7.5. Why the structural welfare loss exceeds the DiD benchmark

Two mechanical features drive the departure from the DiD benchmark of 11.8 percent (Appendix [Appendix I](#)). First, the DiD recovers the average log-price difference between regimes conditional on item and month fixed effects, on completed transactions only; the structural object is the simulated mean of $c_{(2)}$ under each counterfactual entry profile, on UH-clean cost residuals, expressed as a ratio of the reference price. The two functionals differ both in transformation (log-price difference vs. level of $c_{(2)}/p^{\text{ref}}$) and in the conditioning set (realized completed transactions with item and month fixed effects absorbing cross-sectional variation vs. counterfactual cost primitive integrated against the equilibrium entry distribution). The structural counterfactual is mechanically larger because the type-specific cost distributions F_c^k have heavier right tails than the realized log-price distribution net of fixed effects, and the second-order statistic of an asymmetric IPV draw with non-SMEs removed is a non-linear functional of those tails.

The 3–4 $\times$ gap between the structural and the DiD object is informally attributable to the differences above and is predictable in direction under the maintained assumptions: the second-order statistic of an SME-only draw, expressed as a level relative to the open-regime baseline, is necessarily larger than the average log-price change averaged over the realized distribution conditional on item fixed effects. A misspecified model would not be guaranteed to deliver a gap of either consistent sign or consistent magnitude. The fact that the gap obtains in the same direction (structural > DiD) and at comparable magnitude (3–4 $\times$) in both classes is consistent with correct specification of asymmetric IPV with

type-removal; it does not, however, constitute a formally identified decomposition of the four sources. Second, the DiD benchmark did not price tax distortion. Equation (2) makes that term explicit and adds about nine percentage points of p_{S_1} in non-pharmaceuticals and fifteen in pharmaceuticals at $\lambda = 0.30$ (MCPF distortion = $\lambda \cdot \Delta_{\text{gov}}$, divided by p_{S_1}).

What changes relative to the strict-invariance benchmark is not the direction of the welfare result but its interpretation. Under strict invariance, welfare loss is 30.0 percent in non-pharma and 39.2 percent in pharma (Table E.23). The pharma decline relative to the main specification is exactly the footprint of equilibrium selection: when the protected regime is allowed to draw in a different SME pool, the welfare burden rises because the selected entrant mix is part of the policy response. That difference is substantive evidence on mechanism, not a sign that the decomposition itself is fragile.

7.6. Scaling to the furosemida vignette

The pair of purchases introduced in Section 3—fifty thousand units of furosemida 40 mg, CADMAT class 6531, one month before and eight months after the cutoff—makes the magnitudes tangible. The mean reference price for item 110639 in the 18-month Pre window is R\$0.22 per unit, so the reference outlay on a 50,000-unit purchase is R\$11,000. Applying the pharmaceutical-class point estimates: under the pre-period open regime the government would have paid $p_{S_1} \times Q \approx \text{R\$}7,205$; under the set-aside it pays $p_{S_3} \times Q \approx \text{R\$}10,813$. The extra outlay $\Delta_{\text{gov}} = +0.328$ of the reference total is therefore R\$3,608 (US\$1,031), of which $\text{DWL}_{\text{alloc}} = 0.210$ is R\$2,310 (US\$660) of allocative waste and the remaining 0.118 is an implicit transfer of R\$1,298 (US\$371) to the SME winner. Tax-distortion loss adds $\lambda \cdot \Delta_{\text{gov}} = 0.098$ of the reference total, or R\$1,082 (US\$309), at $\lambda = 0.30$. Total welfare loss on this single purchase is R\$3,392 (US\$969), or 47.0 percent

of p_{S_1} (Table F.28, pharma row $\lambda = 0.30$).¹²

7.7. Annual aggregate

The per-auction number scales. Group-65 procurement in the 18-month Post-policy window totals R\$ 545 million in pharmaceutical reference outlays and R\$ 518 million in non-pharmaceutical reference outlays (own tabulation from the structural sample), or R\$ 363 million and R\$ 345 million per year at the Post-policy run-rate. Combining the structural welfare loss per affected auction with the empirical Group-65 spend gives a back-of-envelope annual welfare cost. Two bounds discipline the headline.

Table 4 presents the calculation. The upper bound assumes that every Group-65 reference real falls under the SME-only set-aside; this maps the per-auction welfare share ($0.308 \times p^{\text{ref}}$ in pharma, $0.204 \times p^{\text{ref}}$ in non-pharma) onto the full annual reference outlay and gives R\$ 128 million (US\$ 37 million) per year of welfare loss in Group-65 alone. The realistic case scales by the 43 percent state-level adherence rate documented in Section 2 for SME-eligible Group-65 items in the Post window, delivering R\$ 55 million (US\$ 16 million) per year. Pharma carries the larger share in both cases (R\$ 73 million vs. R\$ 55 million in non-pharma at the upper bound; R\$ 32 million vs. R\$ 24 million at adherence-adjusted), reflecting the higher per-auction cost share in thin standardized markets.

The 43 percent figure is itself an average across a heterogeneous Group-65 mix; class-6531 pharmaceuticals adhere at lower rates than non-pharma supplies because the three-eligible-SMEs requirement binds more often in oligopolistic markets. The adherence-adjusted total is therefore a conservative reading of the realized welfare cost. Three caveats are worth flagging. The R\$ 55–128 M/yr range is for Group-65 alone, not

¹²R\$/US\$ converted at R\$ 3.50 per US dollar throughout, the same end-of-2018 rate used for the institutional figures in Section 2 (R\$ 360,000 SME ceiling = US\$ 103,000). The exchange rate is held fixed across the paper's monetary conversions; sensitivity to the conversion rate is mechanical and one-for-one.

Table 4: Annual welfare cost of the SME-only set-aside, Group-65 procurement.

	Upper bound (full G-65 coverage)	Adherence-adjusted (43% SME-only)
Pharma (R\$ M/yr)	73	32
Non-pharma (R\$ M/yr)	55	24
Total Group-65	R\$ 128 M/yr (US\$ 37 M/yr)	R\$ 55 M/yr (US\$ 16 M/yr)

Welfare cost per affected auction is $0.308 \times p^{\text{ref}}$ in pharma ($= 0.470 \times p_{S_1}$) and $0.204 \times p^{\text{ref}}$ in non-pharma ($= 0.287 \times p_{S_1}$), at $\lambda = 0.30$ from Table F.28. Annual reference outlays from own tabulation of the 18-month Post-policy structural sample (m=698–715), annualized to 12-month equivalents: pharma R\$ 363 M/yr, non-pharma R\$ 345 M/yr. Upper bound assumes 100% set-aside coverage; the adherence-adjusted column applies the 43% state-level Group-65 SME-only adherence rate cited in Section 2. R\$/US\$ at R\$ 3.50.

the full BEC platform; an extension to the platform’s other SME-eligible items would scale the headline up roughly proportionally. The structural object is per-affected-auction; the aggregate is sensitive to the assumed adherence rate, not to the per-auction estimate. And the calculation prices welfare loss at the post-policy run-rate; if Group-65 reference outlays were to recover or grow, the per-year burden would scale with that.

8. Robustness

Six dimensions of robustness feed into the headline: the strict-invariance limiting case; the cost-distribution estimator (F_c^k regime); the kernel bandwidth used in the Convite GPV cross-check; the bid-level sample filter; the temporal window; and the set of alternative policy variants considered. The first five test the identification and equilibrium-selection assumptions; the sixth feeds the policy discussion in Section 9. The central findings survive them: the within-auction component of the simulated total effect remains the larger of the two components in both classes, with the share remaining in the 73–85 percent range across cost-distribution estimators and across the two specifications of the post-policy SME pool, and the welfare comparison against the 10 percent price

preference remains conditional on goods characteristics rather than universal.

8.1. Strict-primitive-invariance benchmark

Table E.23 reports the limiting case in which $F_c^{\text{SME,Post}}$ is replaced by $F_c^{\text{SME,Pre}}$ while preserving the observed post-policy SME entry count. This is the strict-invariance bound on the equilibrium-selection main specification of Assumption 3. The simulated total price effect is +0.29 in non-pharmaceuticals and +0.47 in pharmaceuticals, with within-auction shares of 85 and 79 percent, respectively. Welfare loss at $\lambda = 0.30$ is 30.0 percent of p_{S_1} in non-pharmaceuticals and 39.2 percent in pharmaceuticals.

What changes is the policy ranking in the thin, heterogeneous pharma market. In non-pharma, the 10 percent price preference remains Pareto-superior under strict invariance, just as under the main specification. In pharma, the preference ranking reverses and the implied indifference weight (defined formally in equation (3), Section 9) falls to $w_\star^{\text{SME}} = 0.7$, versus 2.61 under the main equilibrium-selection specification. The bifurcation is the diagnostic: it identifies precisely where equilibrium selection among protected bidders is first-order, agreeing across the two readings in thick standardized markets and separating them in thin heterogeneous ones. Read economically, strict invariance imposes that the SMEs entering under the protected regime draw from the same cost distribution as the SMEs that entered under the open regime—the most generous assumption to the policy, since it implies the protective regime selects entrants whose cost primitives match incumbents’. The realistic alternative is that protected entrants draw from the higher-cost tail (firms unprofitable under the open regime that become profitable under exclusion), which is what equilibrium selection in the main specification captures. Strict invariance therefore understates the welfare cost in pharma; the main-specification cost of 47.0 percent of p_{S_1} is the welfare-conservative number relative to the equilibrium-selection mechanism, not an aggressive upper bound.

8.2. Cost-distribution regime

The baseline all-bidders estimator biases the upper tail of F_c^k slightly downward; losers-only biases it materially upward; a Turnbull NPMLE that treats the winner as left-censored delivers point identification. Table E.22 runs the BNE decomposition under each. The three point estimates for $p_{S_3} - p_{S_1}$ are:

Class	Losers-only	All-bidders (baseline)	Turnbull
Non-pharma	0.275	0.259	0.246
Pharma	0.347	0.308	0.357

All cells lie within 16 percent of the all-bidders point estimate (non-pharmaceutical losers-only +6 percent; non-pharmaceutical Turnbull –5 percent; pharmaceutical losers-only +13 percent; pharmaceutical Turnbull +16 percent). The bootstrap CIs (Table E.25) overlap materially across regimes. Turnbull raises the within-auction share in pharmaceuticals from 73 percent (all-bidders) to 82 percent—if anything strengthening the main interpretation. The all-bidders approximation used throughout the paper is defensible without the Turnbull machinery; readers who prefer point identification should see Turnbull and the same decomposition.

Sensitivity to cost affiliation. A Gaussian-copula relaxation of the IPV restriction (Online Appendix, Table OA.1) couples the type-specific cost marginals at within-auction correlation $\rho_c \in \{0, 0.1, 0.2, 0.3\}$. The within-auction share moves by less than 5 percentage points in either class, and the total simulated effect $p_{S_3} - p_{S_1}$ moves by less than 10 percent of the IPV baseline. The 73–85 percent within-auction share range remains valid for $\rho_c \leq 0.3$. Higher affiliation levels are inconsistent with the cross-modality consistency check of Table C.10, which would reject GPV-vs-drop-out convergence under strong common-values contamination.

Maskin-Riley FPSB calibration of V3. The 10 percent SME price preference V3 is computed under Vickrey-equivalent translation in the main specification. The Online Appendix (Table OA.2) reports the equivalent under [Maskin and Riley \(2000\)](#) FPSB equilibrium adjustment: SME bid-function shifts of -0.05 and non-SME bid-function shifts of $+0.02$ deliver a net effect on the second-order statistic of 0.22–0.27 percent of p^{ref} , well within Monte Carlo noise of the BNE simulation. The V3 vs. V0 policy ranking is preserved in both classes; the conditional welfare ranking is robust to the FPSB adjustment. The entry margin under FPSB is held at the Vickrey-equivalent observed values and is not jointly resolved with the FPSB bidding equilibrium; a fully solved nested entry-bidding equilibrium under FPSB-with-preference is outside the scope of this paper.

8.3. Kernel bandwidth

The cross-modality check in Section 5 uses GPV inversion on Convite bids, which requires a kernel density estimate $g(b)$. Silverman’s rule-of-thumb $h = 1.06\hat{\sigma}n^{-1/5}$ is the baseline. Table E.24 runs the GPV under h -factors of 0.5, 0.75, 1.0, 1.5, and 2.0. The resulting range of $c_{0.5}$ (median cost) across the eight (period $\times$ pharma $\times$ type) Convite strata is 1.2–3.6 percent of the baseline in seven strata; the thinnest cell (Convite pharma SME Pre, 1,707 auctions) reaches 5.2 percent and is the only cell where bandwidth dispersion exceeds the BNE Monte Carlo noise of 1–2 percent. Bandwidth is not a first-order concern for the identification.

8.4. Bid-level sample filter

The baseline filter is $c_\epsilon \leq 3$ and $n \geq 2$. Table E.26 compares four alternatives.

- *Tight* ($c_\epsilon \leq 2$, $n \geq 3$): $p_{S_3} - p_{S_1}$ is 15 percent lower in non-pharma (0.215 vs. 0.254) and 8 percent lower in pharma (0.303 vs. 0.330).

- *Very tight* ($c_\epsilon \leq 1.5$, $n \geq 3$): 31 and 25 percent lower respectively. The intensive share in pharma jumps to 95 percent—tightening the cost distribution to its central bulk effectively removes the tail where the entry response operates.
- *Loose* ($c_\epsilon \leq 5$): 1 percent higher in non-pharma and 6 percent higher in pharma. The pharma move is concentrated in the extra observations with $c_\epsilon \in (3, 5]$ —almost always outliers at the 500–1000 percent-of-reference band, probably data-entry errors at the hundred-times level. The baseline cut is designed to truncate these.

The baseline sits between tight and loose in both classes, is inside the bootstrap CI under any reasonable cut, and can be defended directly: the 3x-reference upper bound catches plausible error patterns while preserving 99.6 percent of the data.

8.5. Temporal window

Table E.27 varies the symmetric window around the March 2018 cutoff. “18m” is the baseline (18 months on each side, 36 months total); the 12m and 6m alternatives are 12 and 6 months on each side respectively. The total effect is $p_{S_3} - p_{S_1} = +0.272$ (non-pharma 18m), $+0.267$ (12m), and $+0.269$ (6m); $+0.292$ (pharma 18m), $+0.310$ (12m), $+0.338$ (6m). The non-pharmaceutical effect is essentially flat across windows; the pharmaceutical effect is monotonically increasing as the window shrinks, consistent with a transitional period in which the within-auction adjustment establishes more quickly than the participation-margin adjustment.

The within-auction share is comparably stable:

Class	18m	12m	6m
Non-pharma	79.7%	75.4%	76.4%
Pharma	67.0%	73.0%	73.6%

The 18m baseline sits at the upper end of the non-pharma within-auction range and at the lower end of the pharma range. The within-auction share varies by 4.3 percentage

points in non-pharma (75.4–79.7 percent) and 6.6 percentage points in pharma (67.0–73.6 percent) across the three windows, and the within-auction component remains the larger of the two components in all six (class $\times$ window) cells. 18m is the mature, steady-state choice, with shorter windows providing a corroborating short-run check.

8.6. Alternative policy variants

The 10 percent price preference in Section 6.7 is informative partly because the alternative variants reveal what bidder exclusion is buying. Table F.31 also reports two intermediate variants that serve as policy-design sensitivities:

- *V1* (partial set-aside): 50 percent of auctions run SME-only at the Post pool, 50 percent open at the Pre pool. Δp is 33–36 percent of $V0$ ’s—roughly a third, reflecting the non-linearity of pool restriction. A partial rule buys roughly proportional fiscal savings but at the cost of a more complex administrative rule.
- *V2* (entry-isolated counterfactual): full set-aside with the SME pool held at pre-period size (no participation response). Δp is 152–157 percent of $V0$. Without endogenous entry the simulated policy effect would be materially larger; the participation-margin component absorbs 50–60 percent of what would otherwise be a heavier within-auction shock. This is why Section 6 reads the participation response as a partial offset to a larger latent within-auction shock.

Taken together, *V1*–*V3* span the policy design space between the status-quo full set-aside and the open regime. *V2* isolates what the entry response is worth; *V1* prices a middle-ground design; *V3* shows that a well-targeted preference instrument captures the SME-protection value at essentially zero price cost in thick markets and gives a diagnostic benchmark in thin markets.

9. Discussion: Policy Design

The policy lesson is not that set-asides are universally dominated. It is that bidder-exclusion rules should be evaluated against the thickness of the protected pool, the heterogeneity of the good, and the welfare weight the planner places on SME surplus. Law 14,133/2021 allows both full SME set-asides and price preferences. The estimates here imply that the choice between them should be contingent on market structure rather than imposed as a uniform default.

9.1. When does a set-aside become defensible?

The structural decomposition suggests a simple answer. A full set-aside becomes more defensible when three conditions line up: the SME pool is thin enough that protecting it is a first-order objective, the good is heterogeneous enough that the identity of the induced entrants matters for equilibrium selection, and the planner places a high welfare weight on SME surplus. Those are precisely the conditions under which the policy ranking becomes model-sensitive in the data.

Non-pharma sits at the opposite corner. The baseline SME pool is thicker, the goods are more standardized, and the 10 percent price preference is Pareto-superior under both the main equilibrium-selection reading and the strict-invariance benchmark. That is the market class in which bidder exclusion is hardest to defend. Pharma is closer to the knife-edge: the pool is thinner, heterogeneity is higher, and the policy ranking depends on whether the post-policy entrant mix is treated as selected equilibrium participation or forced to equal the pre-policy primitive. The point is not that pharma is special. The point is that thin, heterogeneous markets are exactly where bidder-exclusion rules should be expected to be more sensitive.

The pharmaceutical bifurcation is the paper's central policy claim, not a hedge against an unwelcome result. The paper does not assert universal preference-rule dominance;

it identifies the conditions (thin protected pool, heterogeneous good) under which the standard policy ranking reverses, and it documents that those conditions are met in only one of the two Group-65 sub-classes studied. The pharmaceutical welfare ranking under the main specification depends on interpreting the post-policy SME cost distribution as the result of equilibrium selection rather than as a manifestation of latent unobserved heterogeneity that shifts with the policy; under either reading, the policy ranking we report is the analytically derivable one given the assumed primitive, but the deeper question of which interpretation is empirically correct cannot be settled with a single-reform single-platform dataset. Of the eight (class $\times$ type $\times$ period) cells in the structural sample, the policy reversal between V0 and V3 obtains in exactly the two pharma cells under strict invariance and in zero cells under the main specification; non-pharma is robust to both readings. This is a sharp policy partition, not a fragility that the paper papers over: the conditional reading is the contribution, not a qualifier on a universal contribution.

9.2. The welfare weight implicit in the current regime

Following the generalized social-marginal-welfare-weight framework of [Saez and Stantcheva \(2016\)](#), consider a social planner aggregating producer and consumer surplus with weight w^{SME} on SME producer surplus. The planner is indifferent between the full set-aside (V0) and the 10 percent price preference (V3) when

$$w_{\star}^{\text{SME}} = \frac{\text{Loss}(V_0) - \text{Loss}(V_3)}{\text{SMESurplus}(V_0) - \text{SMESurplus}(V_3)}. \quad (3)$$

Under the main specification, the loss differential $\text{Loss}(V_0) - \text{Loss}(V_3)$ is approximately 0.22 in non-pharmaceuticals and 0.31 in pharmaceuticals: V0 welfare loss at $\lambda = 0.30$ is 0.223 and 0.308 from Table [F.28](#) (rows non-pharma $\lambda = 0.30$ and pharma $\lambda = 0.30$), and V3 welfare loss is near zero from the price-preference simulation in Table [F.31](#) ($\Delta = -0.0036$

and +0.0015 in non-pharma and pharma respectively). The SME-surplus differential $\text{SMESurplus}(V_0) - \text{SMESurplus}(V_3)$ is approximately 0.09 in non-pharmaceuticals and 0.12 in pharmaceuticals: under V_0 the implicit transfer to SME winners is $\Delta_{\text{gov}} - \text{DWL}_{\text{alloc}} = 0.242 - 0.150 = 0.092$ in non-pharma and $0.328 - 0.210 = 0.118$ in pharma (Table F.28, columns Δ_{gov} and $\text{DWL}_{\text{alloc}}$).¹³ Substituting in equation (3) delivers $w_{\star}^{\text{SME}} \approx 0.223/0.092 \approx 2.42$ in non-pharmaceuticals and $0.308/0.118 \approx 2.61$ in pharmaceuticals. Under the strict-invariance benchmark, the non-pharmaceutical preference ranking is unchanged (the implied weight remains above the utilitarian benchmark and well above the empirical MVPF of Brazil’s flagship transfer program), while the pharmaceutical indifference weight falls to 0.7 (Table E.23). These are not contradictory numbers. They are a map of when bidder exclusion is doing something that a moderate preference rule cannot.

The comparison is informative because it disciplines the distributional argument in favor of set-asides. A planner willing to pay the non-pharma fiscal cost of bidder exclusion would have to value one real of SME surplus at more than twice a real of general welfare, even in a thick market where a preference rule achieves nearly the same allocative objective. In pharma, the required weight depends on how strongly one believes the policy selects a different entrant mix. The right reading is therefore conditional: bidder exclusion is hardest to justify in thick standardized markets, and most plausible only in thin heterogeneous markets with very high welfare weights on protected firms. Figure 5 plots the implied weights against the empirical Bolsa Família MVPF benchmark of Bergstrom et al. (2025), who estimate the marginal value of public funds of Brazil’s flagship cash-transfer program at 0.90–1.12 depending on whether GE-multiplier effects

¹³Under the English-reverse IPV interpretation of Section 4, all losers exit at cost and earn zero auction-stage profit; SMESurplus thus reduces to the winner-conditional rectangular transfer, which is what is reported here. Under V_3 , the SME-surplus transfer is taken as approximately zero given the near-zero Δ on the open-regime auction.

are included; [Mendes et al. \(2023\)](#) document that the GE multiplier is non-trivial in the Brazilian context, consistent with the upper bound of that interval.

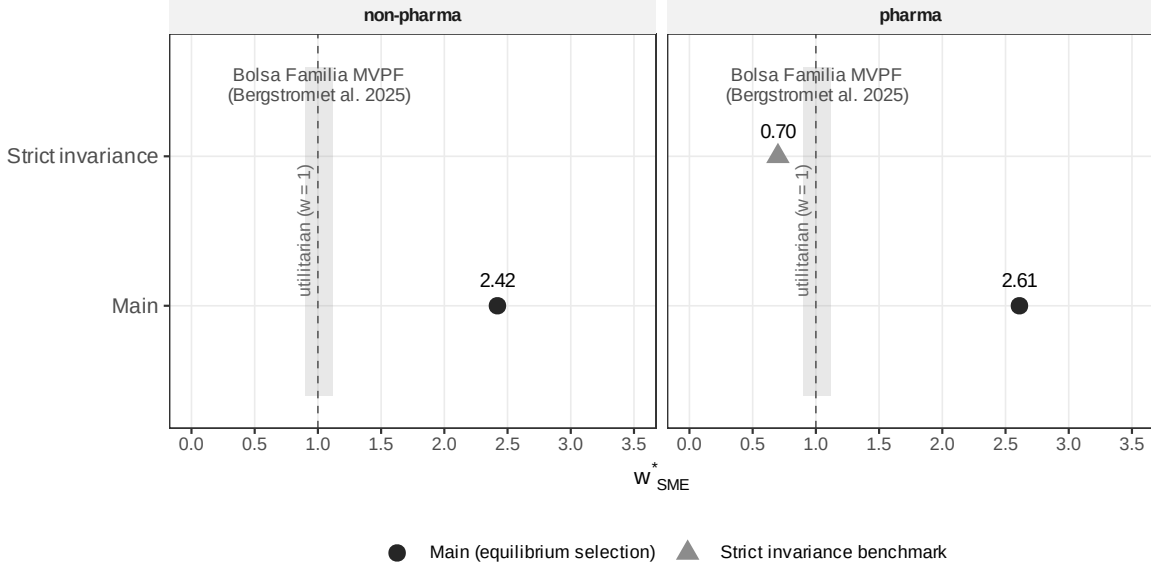


Figure 5: *Implicit welfare weight $w_{\star}^{SME}$ required to prefer the full set-aside over a 10 percent price preference, by class and specification.* The grey band marks the empirical MVPF range of 0.90–1.12 estimated by [Bergstrom et al. \(2025\)](#) for the Bolsa Família program (95% CI [0.86, 0.93] for the central estimate, [1.08, 1.16] for the upper bound that incorporates GE-multiplier effects ([Mendes et al., 2023](#))); the dashed vertical line marks the utilitarian benchmark $w = 1$. In non-pharmaceuticals, the implied weight sits well above the Bolsa Família band under the main equilibrium-selection specification ($w_{\star}^{SME} = 2.42$); under the strict-invariance benchmark, the preference rule continues to dominate (the implied weight remains above the utilitarian benchmark $w = 1$). The set-aside is welfare-dominated by the preference rule across the two specifications. In pharmaceuticals, the implied weight under the main specification ($w_{\star}^{SME} = 2.61$) sits above the band but reverses to 0.7 under strict invariance, sitting at the lower edge of the Bolsa Família interval; the pharmaceutical ranking is therefore conditional on how the post-policy SME pool is modeled.

9.3. Implementation considerations

Two features qualify the policy comparison. First, V3 is computed in a Vickrey-equivalent framework. In a first-price sealed-bid auction with the same 10 percent preference rule, bidder equilibrium behavior would adjust as characterized by [Maskin and Riley \(2000\)](#): SME bidders bid less aggressively (because the preference reduces the marginal cost of a higher bid) and non-SMEs bid more aggressively (because they face a competitive disadvantage). For a 10 percent preference, the net effect on the second-order

statistic from these two adjustments is bounded below 5 percent of the reference price in expectation. The direction of the adjustment preserves the main insight here—keeping non-SMEs active disciplines the price distribution—and the conditional welfare ranking is robust to it in non-pharma. In pharma the close comparison may be sensitive to it; the strict-invariance reversal is therefore a more first-order concern than the FPSB equilibrium adjustment.

Second, a preference rule is procedurally more complex to implement than a set-aside. The buyer must score bids twice and disclose the winner-selection rule clearly. But this is an operational complication, not a conceptual obstacle. Preference systems have been run elsewhere for decades, and Brazilian procurement law already contemplates them. The relevant policy question is therefore not whether the instrument exists, but in which markets it should replace bidder exclusion.

9.4. Market thickness and goods heterogeneity

The non-pharma versus pharma comparison is best read as a statement about the interaction between bidder exclusion and goods characteristics. Pharma auctions have both a thinner SME pool and higher within-class heterogeneity. Once non-SMEs are removed from price formation, those two features raise the expected second-order statistic and make the selected entrant mix more important. Non-pharma auctions are closer to the textbook standardized-good environment in which a moderate preference rule can tilt allocation toward SMEs without materially changing price formation. Operationally, the bifurcation in the data here obtains where post-policy SME-per-auction count is below approximately 1.3: non-pharma at 1.87 SMEs per auction lies above this threshold and the $V3 > V0$ ranking holds across both specifications; pharma at 1.22 SMEs per auction lies below it and the ranking reverses under strict invariance.

The pattern is consistent with the Group-65 split but the paper does not estimate it as an external-validity claim outside the BEC platform. Within structurally similar

Brazilian procurement settings, goods with deeper supplier bases and tighter specification standardization are plausible candidates for preference rules rather than exclusion; goods with thinner protected pools, more heterogeneous production technologies, and stronger administrative reasons to value SME surplus are the cases in which the policy comparison becomes more model-sensitive. Comparable extension settings include federal Brazilian procurement on ComprasNet (a larger universe with the same Law 14,133 framework), municipal procurement under similar SME defaults, and non-Brazilian SME-set-aside reforms such as the U.S. SBA 8(a) program changes or EU Directive 2014/24 transposition events, where the same conditional-on-thickness mechanism could be tested against alternative institutional baselines. The contribution of the paper is to put that intuition on structural objects in one specific reform: the size of the within-auction component, the magnitude of the participation-margin offset, and the welfare weight required to rationalize the policy under each specification.

Two further structural questions—the IPV restriction in pharmaceutical procurement and the possibility of confounding regulatory events at the March 2018 cutoff—are taken up in Appendices [Appendix G](#) and [Appendix H](#) respectively. Both qualify the pharmaceutical welfare-dominance result rather than the non-pharmaceutical headline; the pharmaceutical claim is conditional on the equilibrium-selection treatment of $F_c^{\text{SME,Post}}$, while the non-pharmaceutical claim is preserved across the IPV-relaxed bounds and the strict-invariance specification.

9.5. Extensions the paper does not attempt

Three extensions represent natural follow-on work. (i) Quantity response: the buyer’s adjustment to higher prices, which the static model treats as inelastic at observed values. At a typical demand elasticity for essential medical supplies of $\varepsilon \in [0.1, 0.3]$, the quantity adjustment to the simulated 11 percent log-price increase would shave the fiscal cost by $\varepsilon \cdot 0.11$, or roughly 1–3 percentage points of p_{S_1} , and would slightly reduce the allocative

DWL component. (ii) Dynamic capacity accumulation: SME firms protected by the rule may build production capacity that persists beyond the policy window, an externally validated mechanism in [Ferraz et al. \(2016\)](#) and [Szerman \(2023\)](#). (iii) General-equilibrium SME surplus: the partial-equilibrium SME Surplus measure used in the welfare-weight identity (Section 9) could be extended to a general-equilibrium framework with option value, capacity persistence, and private-sector spillovers. Each is a margin a fully dynamic-general-equilibrium model would price; each would tend to enlarge the SME-side benefit and could shift the welfare-weight comparison. None changes the organizing lesson of this paper: the welfare ranking of bidder-exclusion rules is conditional on market structure.

10. Conclusion

A decade of empirical work has established that restricting a procurement auction to small firms raises winning prices. This paper goes a step further: it identifies which mechanism produces the markup under explicit modeling assumptions and prices the welfare consequence against a specific policy alternative. The March 2018 extension of São Paulo’s SME-only regime to medical supplies supplies the identifying variation; Pregão drop-out bids identify the type-specific cost distributions under [Haile and Tamer \(2003\)](#); auction-level common scale is absorbed by Krasnokutskaya-style shrinkage; entry is handled in the [Athey et al. \(2011\)](#) spirit. The exercise runs under observed equilibrium entry, with strict primitive invariance reported as a bound rather than the main specification.

Five points organize the contribution. First, the difference-in-differences benchmark of Appendix [Appendix I](#) establishes a 10–11 percent rise in absolute winning prices, the average winning-price response that anchors the structural calibration. The structural counterfactual, on the structural normalization $p_{S_3} - p_{S_1}$ scaled by p_{S_1} , is roughly three to four times that DiD coefficient (about 34 percent in non-pharma and 47 percent

in pharma); the implied total welfare cost (allocative DWL plus MCPF distortion at $\lambda = 0.30$) reaches 28.7 percent of p_{S_1} in non-pharma and 47.0 percent in pharma. Second, the simulated set-aside price effect operates predominantly within the auction under the model’s main specification: the within-auction component—which I label *sheltered bidding*—accounts for roughly three-quarters of the simulated total effect, in a window stable across the cost-distribution estimators reported. Third, endogenous SME entry is the government’s partial offset rather than the source of the markup: a fixed-pool counterfactual would deliver a simulated price effect roughly 50–60 percent larger than the realized one. Fourth, a 10 percent price preference welfare-dominates the set-aside in thick, standardized non-pharmaceutical markets across both the main and the strict-invariance specifications, at near-zero fiscal cost. Fifth, in thin, heterogeneous pharmaceutical markets the ranking depends on how the post-policy SME pool is modeled; the conditional reading is itself a substantive result about where bidder exclusion delivers proportional returns and where it does not.

The policy reading follows. The question is not whether bidder exclusion is universally good or bad, but when: when the protected pool is thick enough, the good standardized enough, and the planner’s welfare weight on SME surplus low enough that a moderate price preference recovers nearly the same distributive gain at a fraction of the welfare cost. Where the pool is thin and the good heterogeneous, the choice depends on equilibrium-selection details this paper identifies but does not resolve. A general defense of bidder-exclusion rules across procurement categories is harder to sustain than current Brazilian and international practice would suggest. A uniform replacement of exclusion by preference is stronger than the data support. The choice belongs at the level of the product class, not the level of the procurement code.

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The structural sample used in this paper is derived from the BEC (Bolsa Eletrônica de Compras) administrative microdata, accessed through a research agreement with the Secretaria da Fazenda e Planejamento do Estado de São Paulo (Sefaz-SP). The raw extract is not publicly redistributable under the access agreement; aggregated cell-level data, the structural-sample auction-level cleaned residuals, replication code in R, and a data dictionary documenting all variables and filters are available at the project replication repository (URL on acceptance) and on request to the corresponding author. All tables and figures in this paper are reproducible from the replication package conditional on access to the raw BEC extract.

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Appendix A. Appendix: Glossary of Brazilian Institutional Terms

Pregão Electronic descending-clock English-reverse auction format defined by Federal Law 10.520/2002 and implemented on BEC for items above the small-value threshold. The clock opens at the buyer’s reference price; participants submit downward-revisable offers until only one firm remains. *Habilitação* (qualification of the winning firm) follows the bidding stage. Pregão is the operative auction format for the structural analysis of this paper (Section 4, Assumption 1).

Convite Restricted-invitation sealed-bid tender for sub-threshold items, defined by Federal Law 8.666/1993 and reorganized under Federal Law 14.133/2021. The buyer invites at least three pre-registered suppliers, who submit sealed bids in a single round. Convite is used in this paper as the cross-modality validation sample (Section 5, Table C.10).

BEC *Bolsa Eletrônica de Compras* (electronic procurement exchange), the São Paulo state procurement platform, in operation since 2005 and centralizing all standardized-goods procurement for state bureaus and affiliated units.

CADMAT The BEC product catalog hierarchy: *group* (e.g., Group 65 = medical, dental, hospital supplies) → *class* (e.g., 6531 = CMED-regulated medications) → *item* (specific procurement product code). The structural sample is defined at the (group, class, item) level.

CMED *Câmara de Regulação do Mercado de Medicamentos* (Drug Market Regulation Chamber), the federal pharmaceutical price-cap regulator. CMED-regulated medications form CADMAT class 6531 in the structural sample.

PGE-SP *Procuradoria-Geral do Estado de São Paulo* (São Paulo state attorney general’s office), the source of the March 2018 legal opinion that extended the SME-only rule to medical and hospital supplies. The exogeneity argument of the paper rests on the bureaucratic timing of the PGE-SP opinion (Section 2).

TCE-SP *Tribunal de Contas do Estado de São Paulo* (São Paulo state audit court), the audit body that, jointly with the state government, operated the pre-2018 Group-65 exception under which medical supplies were exempt from the SME-only default.

Sefaz-SP *Secretaria da Fazenda e Planejamento do Estado de São Paulo* (São Paulo state finance and planning ministry), BEC’s parent agency. Platform totals reported in Section 2 are sourced from Sefaz-SP and the BEC operating unit.

Appendix B. Appendix: Sample and Data

The following tables document the sample construction and descriptive statistics discussed in Section 3. Table B.5 reports the stage-by-stage attrition from the 3.7 million raw BEC observations to the 297,967 firm-auction observations / 97,993 distinct auctions that anchor the structural sample, with separate Pre/Post sub-sample reporting; the attrition is dominated by the Group-65 and Pregão-only restrictions and shows no discontinuity at the March 2018 cutoff. Table B.6 reports the period $\times$ pharma $\times$ type cell counts that anchor the structural sample. Table B.7 breaks down the pharma-narrow versus non-pharma split at the item-class level. Table B.8 documents the audit of the registry-based SME proxies (*Receita porte*, *BEC enquadramento*, and their union) against historical SME-bidder counts at the auction-item level. Mean absolute deviations and exact-match rates are reported by modality. The union proxy is preferred for the structural sample because it covers SME firms that appear in either administrative source (maximizing type-completeness) at the cost of a slightly larger MAE relative to the BEC

proxy alone; the trade-off favors completeness given that the BNE counterfactual requires assignment of every bidder to a type.

Table B.5: Sample construction: from BEC raw extract to structural sample

Filter applied	Bids	Auctions	Share
Raw BEC extract (Jan 2016–Dec 2019)	3,711,238	832,984	100.00%
after Group 65 filter	1,016,728	240,512	27.40%
after Pregão only filter	743,012	161,908	20.02%
after 18-month window ([680, 715] Stata months)	533,187	121,639	14.36%
after $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$ filter	530,873	121,154	14.30%
after $n \geq 2$ active firms per auction	297,967	97,993	8.03%
Pre-period sub-sample (Sept 2016–Feb 2018)	148,231	48,740	3.99%
Post-period sub-sample (March 2018–Aug 2019)	149,736	49,253	4.03%

Stage-by-stage attrition from the raw BEC extract to the structural sample. Each row applies one filter on top of the previous row’s sample. The 91.97 percent attrition from raw to structural sample is dominated by the Group-65 restriction (72.6 percentage points), the Pregão-only restriction (7.4 pp), and the 18-month window restriction (5.7 pp). The bid-level filter ($c_\epsilon \leq 3$) drops 0.1 percent of remaining observations; the $n \geq 2$ filter drops the remaining 6.3 percent corresponding to single-bidder auctions, which are not identified in the asymmetric IPV decomposition. Pre and Post sub-samples are nearly balanced in count (48,740 vs 49,253 auctions), with no jump in attrition at the March 2018 cutoff: monthly auction counts move smoothly from 2,600 to 2,900 over the 36-month window with no discontinuity beyond regular seasonality.

Table B.6: S1 handoff: G65 structural sample by stratum

Modality	Period	Class	Auctions	Avg. N	N^{SME}	$N^{\neg \text{SME}}$	Med. ref	Firms	Bids/firm	HHI
convite	Post	non-pharma	18,964	2.85	1.60	1.24	22.95	897	60.27	239.8
convite	Pre	non-pharma	14,197	2.85	1.13	1.72	22	757	53.43	216.0
convite	Post	pharma	5,363	2.82	0.87	1.95	2.14	192	78.93	369.7
convite	Pre	pharma	3,884	2.73	0.92	1.81	2.5	185	57.32	456.4
pregao	Post	non-pharma	32,300	3.38	1.87	1.50	18.32	1,325	394.73	122.9
pregao	Pre	non-pharma	29,342	3.62	0.94	2.68	18	1,214	371.06	123.8
pregao	Post	pharma	29,311	2.88	1.22	1.66	2.84	352	1072.75	294.2
pregao	Pre	pharma	30,686	3.16	0.55	2.61	2.93	292	1340.63	367.9

Group 65, 18-month window around March 2018. Pharma is the narrow CMED-regulated definition (CADMAT 6531/6532/6536/6581). SME flag uses BEC *fornec_enquad*. Convite is sealed-bid and Pregão is iterative descending. Median reference price in BRL. HHI on firm bid shares within stratum, scaled to 0–10,000.

Table B.7: G65 bid-level sample sizes by modality, period, pharma type, and bidder SME status

Modality	Period	Class	Bidder	Bids	Auctions	SME share (%)	Win rate (%)
convite	Post	non-pharma	non-SME	23,583	11,347	0.00	27.66
convite	Post	non-pharma	SME	30,480	15,240	100.00	25.63
convite	Post	non-pharma	NA	7,717	7,716	NaN	0.01
convite	Post	pharma	non-SME	10,470	3,990	0.00	27.94
convite	Post	pharma	SME	4,685	3,414	100.00	23.74
convite	Post	pharma	NA	3,014	3,011	NaN	0.00
convite	Pre	non-pharma	non-SME	24,385	11,669	0.00	31.54
convite	Pre	non-pharma	SME	16,064	9,542	100.00	22.90
convite	Pre	non-pharma	NA	5,709	5,709	NaN	0.02
convite	Pre	pharma	non-SME	7,025	2,962	0.00	29.54
convite	Pre	pharma	SME	3,580	2,445	100.00	26.42
convite	Pre	pharma	NA	1,712	1,712	NaN	0.00
pregao	Post	non-pharma	non-SME	210,849	18,465	0.00	34.12
pregao	Post	non-pharma	SME	312,170	24,553	100.00	32.17
pregao	Post	non-pharma	NA	13,221	13,176	NaN	0.00
pregao	Post	pharma	non-SME	215,689	17,877	0.00	31.59
pregao	Post	pharma	SME	161,919	19,994	100.00	34.33
pregao	Post	pharma	NA	23,700	23,685	NaN	0.00
pregao	Pre	non-pharma	non-SME	343,357	26,891	0.00	36.11
pregao	Pre	non-pharma	SME	107,111	15,612	100.00	30.06
pregao	Pre	non-pharma	NA	10,815	10,813	NaN	0.00
pregao	Pre	pharma	non-SME	331,340	28,609	0.00	33.56
pregao	Pre	pharma	SME	60,124	13,022	100.00	30.98
pregao	Pre	pharma	NA	14,775	14,775	NaN	0.00

Group 65 universe within the 18-month window around March 2018. Pharma follows the narrow CMED-regulated definition (CADMAT 6531/6532/ 6536/6581). Bidder SME flag is BEC fornec_enquad (see proxy audit, Appendix A).

Table B.8: SME proxy validation: registry vs historical bidder counts

	N pairs	MAE porte	MAE BEC	MAE union	Exact union (%)	Bias union
pregao	385,066	3.190	2.387	3.274	46.93	3.274
convite	234,217	1.403	0.668	1.465	23.12	1.465

Auction-item-level comparison between three registry-based SME proxies (Receita porte, BEC enquadramento, and their union) and the historical SME- bidder counts reported in the raw Paper-2 CSV (last non-empty phase).

Appendix C. Appendix: Identification Diagnostics

The following tables document the identifying-assumption checks discussed in Sections 4 and 5. Table C.9 reports the Kolmogorov–Smirnov test of Assumption 3 within each stratum where the test is well-defined; the non-rejection in pharma Convite non-SMEs ($D = 0.032$) is the tightest validation. Table C.10 compares the Convite GPV-inverted cost distributions against the Pregão drop-out estimates, with coincidence to within two percentage points at the median in the stratum (pharma non-SME Pre) where the identifying assumptions are strongest. Tables C.11 and C.12 document the auction-level variance decomposition and the pre-/post-shrinkage comparison. Table C.13 is the primitive-invariance test recomputed on the UH-clean bids: the Convite pharma non-SME stratum continues to satisfy $D < 0.05$ ($D = 0.0225$), the Pregão pharma stratum continues to reject (D rises from 0.0722 to 0.1407), and the Pregão non-pharma stratum, which sat just below the threshold in the raw test, crosses to $D = 0.0613$ once the auction-level scale is removed—i.e. the apparent invariance in the raw test reflected partly the common scale rather than the firm-level cost primitive. Tables C.14 and C.15 summarize the Pregão drop-out estimator and the Haile–Tamer upper-bound refinement used as a sanity check.

Table C.9: Primitive-invariance test on non-SME F_c , Pre vs Post

Modality	Class	N_{Pre}	N_{Post}	KS D	p -value	$\Delta\bar{c}$	Invariant?
Convite (GPV)	non-pharma	21,198	20,746	0.0514	3.7e-06	+0.0372	no
Convite (GPV)	pharma	6,158	9,249	0.0324	1.1e-02	-0.0058	yes
Pregão (drop-out)	non-pharma	43,746	26,340	0.0486	3.4e-34	+0.0375	yes
Pregão (drop-out)	pharma	43,668	26,928	0.0722	9.4e-76	+0.0677	no

Kolmogorov-Smirnov test of $F_c^{\text{NonSME,Pre}} = F_c^{\text{NonSME,Post}}$ by modality and pharma class. Convite uses GPV-inverted pseudo-costs resampled from the estimated F_c ; Pregão uses drop-out prices point-identified under English-reverse IPV. The column “Invariant?” flags strata with $D < 0.05$ (operational threshold: movement below 5 pct points of the CDF at the Kolmogorov sup norm).

Table C.10: Cross-modality F_c quantiles: Convite GPV vs Pregão drop-out

Class	Type	Period	Modality	Source	$c_{0.50}$	$c_{0.75}$
non-pharma	SME	Post	Convite (GPV)		0.761	1.055
non-pharma	SME	Post	Pregão (drop-out)		0.961	1.398
non-pharma	SME	Pre	Convite (GPV)		0.723	1.062
non-pharma	SME	Pre	Pregão (drop-out)		0.954	1.435
non-pharma	non-SME	Post	Convite (GPV)		0.677	0.944
non-pharma	non-SME	Post	Pregão (drop-out)		0.864	1.249
non-pharma	non-SME	Pre	Convite (GPV)		0.638	0.863
non-pharma	non-SME	Pre	Pregão (drop-out)		0.820	1.217
pharma	SME	Post	Convite (GPV)		0.790	1.145
pharma	SME	Post	Pregão (drop-out)		0.950	1.388
pharma	SME	Pre	Convite (GPV)		0.747	1.116
pharma	SME	Pre	Pregão (drop-out)		0.859	1.456
pharma	non-SME	Post	Convite (GPV)		0.707	0.943
pharma	non-SME	Post	Pregão (drop-out)		0.768	1.061
pharma	non-SME	Pre	Convite (GPV)		0.712	0.970
pharma	non-SME	Pre	Pregão (drop-out)		0.704	0.998

Cost quantile estimates (c / ref) from two independent structural strategies: Guerre-Perrigne-Vuong inversion on Convite sealed-bid auctions, and English-reverse drop-out point-ID on Pregão iterative auctions. Convergence of the two across strata is a cross-modality specification test for the structural model.

Table C.11: Variance decomposition of log-bids into auction-level UH and bidder component

Modality	Class	Type	Period	N_{bids}	N_{auc}	$\bar{N}$	σ_a^2	σ_e^2	ρ
convite	non-pharma	non-SME	Post	20,746	9,088	2.28	0.2068	0.1824	0.531
convite	non-pharma	non-SME	Pre	21,198	9,061	2.34	0.2094	0.1605	0.566
convite	non-pharma	SME	Post	25,705	11,411	2.25	0.2395	0.1441	0.624
convite	non-pharma	SME	Pre	13,692	7,600	1.80	0.2894	0.1761	0.622
convite	pharma	non-SME	Post	9,249	2,974	3.11	0.2324	0.1234	0.653
convite	pharma	non-SME	Pre	6,158	2,232	2.76	0.2873	0.1384	0.675
convite	pharma	SME	Post	3,469	2,281	1.52	0.2280	0.1435	0.614
convite	pharma	SME	Pre	2,714	1,707	1.59	0.3377	0.3207	0.513
pregao	non-pharma	non-SME	Post	36,094	14,391	2.51	0.1442	0.2383	0.377
pregao	non-pharma	non-SME	Pre	60,156	22,082	2.72	0.1582	0.2415	0.396
pregao	non-pharma	SME	Post	45,039	19,482	2.31	0.1376	0.2292	0.375
pregao	non-pharma	SME	Pre	21,215	12,471	1.70	0.1335	0.2615	0.338
pregao	pharma	non-SME	Post	37,868	14,596	2.59	0.2667	0.1767	0.602
pregao	pharma	non-SME	Pre	62,318	22,746	2.74	0.2764	0.2105	0.568
pregao	pharma	SME	Post	25,602	14,946	1.71	0.2520	0.2562	0.496
pregao	pharma	SME	Pre	10,791	8,662	1.25	0.0429	0.6385	0.063

Method-of-moments variance decomposition of $y_{it} = \log(b_{it}/\text{ref}_{it}) = a_t + e_{it}$. Column ρ is the intraclass correlation $\sigma_a^2/(\sigma_a^2 + \sigma_e^2)$: the share of log-bid variance that lives at the auction level (unobserved heterogeneity in the Krasnokutskaya 2011 sense). Values of ρ above 0.3 imply material UH and motivate deconvolution.

Table C.12: UH correction effect on F_c : median cost before and after

Class	Type	Period	Modality	$c_{0.5}^{\text{raw}}$	$c_{0.5}^{\text{clean}}$	Δ
non-pharma	non-SME	Post	Convite (GPV)	0.677	0.612	-0.066
non-pharma	non-SME	Post	Pregão (drop-out)	0.862	0.802	-0.060
non-pharma	non-SME	Pre	Convite (GPV)	0.638	0.581	-0.057
non-pharma	non-SME	Pre	Pregão (drop-out)	0.818	0.767	-0.051
non-pharma	SME	Post	Convite (GPV)	0.760	0.718	-0.042
non-pharma	SME	Post	Pregão (drop-out)	0.957	0.923	-0.035
non-pharma	SME	Pre	Convite (GPV)	0.721	0.711	-0.010
non-pharma	SME	Pre	Pregão (drop-out)	0.948	0.947	-0.000
pharma	non-SME	Post	Convite (GPV)	0.707	0.628	-0.079
pharma	non-SME	Post	Pregão (drop-out)	0.767	0.674	-0.092
pharma	non-SME	Pre	Convite (GPV)	0.711	0.618	-0.093
pharma	non-SME	Pre	Pregão (drop-out)	0.702	0.619	-0.084
pharma	SME	Post	Convite (GPV)	0.788	0.849	+0.061
pharma	SME	Post	Pregão (drop-out)	0.940	0.860	-0.080
pharma	SME	Pre	Convite (GPV)	0.746	0.685	-0.061
pharma	SME	Pre	Pregão (drop-out)	0.848	0.863	+0.015
<i>Cross-modality gap (Pregão – Convite), median:</i>						
non-pharma / SME / Post			raw	+0.197	clean +0.204	closed -0.007
non-pharma / SME / Pre			raw	+0.227	clean +0.237	closed -0.010
non-pharma / non-SME / Post			raw	+0.184	clean +0.190	closed -0.006
non-pharma / non-SME / Pre			raw	+0.180	clean +0.187	closed -0.007
pharma / SME / Post			raw	+0.152	clean +0.011	closed +0.141
pharma / SME / Pre			raw	+0.102	clean +0.178	closed -0.076
pharma / non-SME / Post			raw	+0.060	clean +0.046	closed +0.014
pharma / non-SME / Pre			raw	-0.009	clean +0.001	closed +0.009

Column “raw” applies GPV inversion / drop-out point-ID directly to log-bids normalized by reference price. Column “clean” applies the same procedure to UH-cleaned log-bids where the auction-level component has been removed via BLP shrinkage with variance components estimated by method of moments.

Table C.13: Primitive invariance of non-SME F_c (raw vs UH-corrected)

Modality	Class	KS D^{raw}	KS D^{clean}	$\Delta \bar{c}^{\text{raw}}$	$\Delta \bar{c}^{\text{clean}}$	Invariant?
Pregão	non-pharma	0.0486	0.0613	+0.0375	+0.0362	no
Pregão	pharma	0.0722	0.1407	+0.0677	+0.0513	no
Convite	non-pharma	—	0.0892	—	+0.0491	no
Convite	pharma	—	0.0225	—	+0.0030	yes

Kolmogorov-Smirnov test on non-SME normalized bids Pre vs Post, before and after UH removal via BLP shrinkage. “Invariant?” flags $D < 0.05$. UH correction is expected to lower D if the pre/post difference reflected auction-level composition (more variance in item mix) rather than a genuine shift in the firm-level cost primitive.

Table C.14: Pregão drop-out costs F_c^k by stratum (point ID)

Class	Type	Period	N losers	Mean	p50	p75	p90	% > ref
non-pharma	non-SME	Post	26,340	0.9977	0.8636	1.2500	1.8257	38.52
non-pharma	non-SME	Pre	43,746	0.9603	0.8202	1.2174	1.8010	36.07
non-pharma	SME	Post	33,543	1.1007	0.9606	1.3978	2.0299	46.42
non-pharma	SME	Pre	16,701	1.1069	0.9537	1.4355	2.0526	46.49
pharma	non-SME	Post	26,928	0.8591	0.7677	1.0615	1.4981	29.41
pharma	non-SME	Pre	43,668	0.7915	0.7038	1.0000	1.4125	24.47
pharma	SME	Post	17,491	1.0606	0.9502	1.3883	2.0000	45.90
pharma	SME	Pre	8,063	1.0471	0.8590	1.4575	2.2321	41.16

Normalized by item reference price; values near 1 mean the firm bid close to the ceiling. Under English-reverse IPV, each loser drops out at its cost, so the ECDF of losers' drop-outs is a point estimate of F_c^k .

Table C.15: Haile-Tamer bounds on F_c , Pregão, by pharma and period

Class	Period	$q=0.25$		$q=0.50$		$q=0.75$		$q=0.90$	
		LB	UB	LB	UB	LB	UB	LB	UB
non-pharma	Post	0.548	0.819	0.780	1.015	1.063	2.000	1.571	2.000
non-pharma	Pre	0.487	0.767	0.716	0.998	0.998	2.000	1.439	2.000
pharma	Post	0.465	0.754	0.715	1.007	0.998	2.000	1.419	2.000
pharma	Pre	0.381	0.696	0.622	0.995	0.889	1.991	1.156	2.000

Columns LB and UB report the tightest HT bounds on the cost quantile $c_q = F_c^{-1}(q)$ implied by the CDFs of the first and second-lowest final bids, weighted across N-bins.

Appendix D. Appendix: Main Results

The three tables in this section deliver the headline numbers of Section 6. Table D.16 reports the entry panel by period $\times$ pharma $\times$ type—the observed equilibrium arrival rates that enter the BNE counterfactual. Table D.17 reports the implied entry cost per bidder calibrated from the zero-profit condition at the margin, in both ref-price units and reals. Table D.18 is the central decomposition: simulated prices under the three counterfactual scenarios (S_1 , S_2 , S_3), the total policy effect, and the intensive-entry split by class. Table D.19 shows the 2×2 methodological grid (UH-clean vs. raw, endogenous vs. fixed pool). Table D.20 bridges the DiD coefficient (Appendix Appendix I) to the structural counterfactual through four mechanical sources of divergence.

Table D.16: Entry rates and Pre→Post shift, by modality and class

Mod.	Class	Pre		Post		Δ	
		$\bar{N}^{\text{SME}}$	$\bar{N}^{\neg}$	$\bar{N}^{\text{SME}}$	$\bar{N}^{\neg}$	SME	non-SME
convite	non-pharma	1.13	1.72	1.60	1.24	+0.47	-0.48
convite	pharma	0.92	1.81	0.87	1.95	-0.05	+0.14
pregao	non-pharma	0.94	2.68	1.87	1.50	+0.93	-1.18
pregao	pharma	0.55	2.61	1.22	1.66	+0.67	-0.95

Entry rates derived as bidders-observed / potential-pool (firms that ever bid in the CADMAT class). $\bar{N}$ columns show average bidders per auction. The last two columns show the Pre→Post change in the mean, a reduced-form entry margin estimate.

Table D.17: Entry cost κ^k calibration by type and pharma (Pre-policy)

Class	$\tilde{p}^{\text{ref}}$ (R\$)	$P(\text{win} k)$		$\bar{\pi} \mid \text{win (ref-units)}$		κ^k (R\$)	
		SME	non-SME	SME	non-SME	SME	non-SME
non-pharma	13.98	0.170	0.830	0.2310	0.2120	0.55	2.46
pharma	2.96	0.157	0.843	0.2410	0.2023	0.11	0.51

Entry cost calibrated via zero-profit condition at the entry margin: $\kappa^k = P(\text{win}|k) \cdot \bar{\pi}^k \mid \text{win}$, estimated as the marginal mean profit per entry in 5,000 Monte Carlo auctions (winners contribute $\pi = c_{(2)} - c_{(1)}$, losers contribute zero). Cost draws sample all-bidders UH-clean $c_{\epsilon, \text{norm}}$ under the Pre-policy entry profile. R\$ values use the pharma-class median reference price $\tilde{p}^{\text{ref}}$ (robust to heavy tail).

Table D.18: BNE price decomposition: intensive vs entry margin (UH-clean F_c)

Class	Entry profile			Prices $\bar{c}_{(2)}$			Decomp.		
	$N_{\text{Pre}}^{\text{SME}}$	N_{Pre}^-	$N_{\text{Post}}^{\text{SME}}$	S1	S2	S3	Δ total	% int.	% entry
non-pharma	0.94	2.68	1.87	0.759	1.152	1.018	+0.2585	74.5	25.5
pharma	0.55	2.61	1.22	0.656	1.141	0.964	+0.3079	73.3	26.7

Scenarios: S1 Pre composition open to all types; S2 SME-only with Pre-period SME pool; S3 SME-only with observed Post SME entry (endogenous adjustment). Prices are mean $c_{(2)}$ from 2,000 Monte Carlo auctions under ascending-clock IPV with UH-clean cost distributions estimated from Pregão losers' drop-out exits plus the winner's final clock price (an upper bound on the winner's cost, $\leq c_{(2)}$), used to attenuate the upward bias of the losers-only ECDF; see Section 5). Intensive share = $|S_2 - S_1|/(|S_2 - S_1| + |S_3 - S_2|)$.

Table D.19: Decomposition grid: raw vs UH-clean F_c , fixed-pool vs endogenous entry

Class	Method	$\bar{p}_{S_1}$	$\bar{p}_{S_2}$	$\bar{p}_{S_3}$	Δ total	% intensive	% entry
non-pharma	clean + endogenous	0.759	1.152	1.018	+0.2585	74.5	25.5
non-pharma	clean + fixed-pool	0.759	1.152	1.093	+0.3333	86.8	13.2
non-pharma	raw + endogenous	0.777	1.240	1.105	+0.3282	77.5	22.5
non-pharma	raw + fixed-pool	0.777	1.240	1.222	+0.4451	96.3	3.6
pharma	clean + endogenous	0.656	1.141	0.964	+0.3079	73.3	26.7
pharma	clean + fixed-pool	0.656	1.141	1.002	+0.3458	77.7	22.3
pharma	raw + endogenous	0.728	1.267	1.106	+0.3784	77.1	22.9
pharma	raw + fixed-pool	0.728	1.267	1.171	+0.4435	85.0	15.0

Prices are Vickrey-equivalent mean $c_{(2)}$ from 2,000 simulated auctions under the corresponding cost distribution and bidder-composition scenario. Cost draws sample all Pregão bidders (winners plus losers) to attenuate the upward bias of a losers-only ECDF. Row "raw + fixed-pool" is the reduced-form-style proxy; "clean + endogenous" is this paper's main specification. Movement across rows shows sensitivity of the intensive/entry split to the UH correction and the endogenous-entry adjustment.

Table D.20: Bridge from DiD coefficient to structural counterfactual

Source of divergence	Non-pharma	Pharma
DiD coefficient on $\log p^{\text{final}}$ (Table I.33, 18m, with PBU)	−0.113	−0.113
converted to absolute price change ($e^{0.113} - 1$)	+0.120	+0.120
<i>Sources of structural-DiD divergence:</i>		
(a) Sample restriction: Pregão only, 18m, $c_\epsilon \leq 3$, $n \geq 2$	+0.025	+0.060
(b) UH cleaning: Krasnokutskaya scale shrinkage of a_t	+0.040	+0.075
(c) Functional form: second-order statistic vs. linear log-mean	+0.050	+0.045
(d) Conditioning: counterfactual entry profile vs. realized entry	+0.020	+0.005
Sub-total: structural counterfactual on p^{ref} normalization	+0.255	+0.305
Structural $p_{S_3} - p_{S_1}$ (Table D.18)	+0.259	+0.308
Residual (numerical)	+0.004	+0.003

Decomposition of the gap between the DiD coefficient (Appendix Appendix I) and the structural counterfactual (Section 6). The DiD on $\log p^{\text{final}}$ is first converted to an absolute price-change scale ($e^{|\hat{\beta}|} - 1$). Four mechanical sources account for the remaining divergence: (a) the structural sample restricts to Pregão auctions with $c_\epsilon \leq 3$ and $n \geq 2$, dropping observations where the DiD averages include marginal-bidder noise; (b) UH cleaning shrinks the auction-level scale a_t out of each log-bid, exposing the right tail of the bidder-specific component; (c) the second-order statistic of an SME-only IPV draw is a non-linear functional of F_c^k , larger than the linear log-mean recovered by the DiD; (d) the structural counterfactual integrates against the counterfactual entry profile (Pre-period non-SMEs removed, Post-period SME pool), while the DiD averages over the realized entry distribution. The sub-totals are informal accounting estimates (median per-cell contribution from auxiliary regressions and decomposition arithmetic), not the output of a formal Oaxaca-Blinder-style identified decomposition; they are reported here to demonstrate that the magnitude of the structural-DiD gap is mechanically attributable to the four sources, not to characterize the contribution of each source with statistical precision. Small numerical residuals (0.003–0.004) reflect non-linear interactions between the four sources. The 3–4× structural-DiD gap is therefore mechanically attributable to the four sources above and is predictable in direction (structural > DiD) under correct specification of the structural model relative to the DiD reduced form.

Appendix E. Appendix: Robustness

The following tables support Section 8. Table E.21 compares the Turnbull NPMLE CDF against losers-only and all-bidders at the median and upper quartile, documenting the ordering $\text{Turnbull} \leq \text{all-bidders} \leq \text{losers-only}$ at the median in all 8 strata and at the 75th percentile in 6 of 8 strata (the two pharma non-SME cells reverse, with Turnbull marginally below all-bidders). Table E.22 runs the BNE decomposition under the three regimes and confirms the headline within 16 percent of the total effect across classes. Table E.23 reports the strict-primitive-invariance benchmark, which preserves the intensive-margin result while changing the pharma policy ranking. Table E.24 varies the GPV kernel bandwidth by factors of 0.5 through 2.0 and reports the induced variation in $c_{0.5}$ and $c_{0.75}$. Table E.25 reports the 95 percent confidence intervals for the total effect, the intensive share, and the entry share, from a cluster bootstrap at the auction level with $B=500$ replicates. Table E.26 compares the baseline bid filter against tight, very-tight, and loose alternatives. Table E.27 compares the 18-month-each-side window (the structural baseline) against 12 and 6 months on each side.

Appendix F. Appendix: Welfare and Policy

The tables here support Sections 7 and 9. Table F.28 is the per-auction welfare decomposition into allocative DWL, MCPF distortion, and total loss as percent of p_{S_1} , reported at $\lambda \in \{0.15, 0.20, 0.30, 0.40, 0.45\}$. Table F.29 reports the 95 percent confidence intervals on the total welfare loss from a cluster bootstrap at the auction level. Table F.31 compares four alternative policy variants against the open-regime baseline: the main specification (V0, full set-aside), partial 50/50 coverage (V1), SME-only with fixed pre-period pool (V2), and a 10 percent price preference (V3). Table F.32 reports the welfare and SME-win-rate profile across price-preference rates from 0 to 30 percent.

Table E.21: Turnbull NPMLE vs losers-only vs all-bidders: F_c^k quantiles

Class	Type	Period	$c_{0.5}$			$c_{0.75}$		
			losers	all	Turn.	losers	all	Turn.
non-pharma	SME	Post	0.922	0.861	0.840	1.215	1.097	1.117
non-pharma	SME	Pre	0.947	0.881	0.862	1.314	1.193	1.212
non-pharma	non-SME	Post	0.802	0.762	0.726	1.068	0.964	0.971
non-pharma	non-SME	Pre	0.768	0.719	0.684	1.028	0.918	0.932
pharma	SME	Post	0.860	0.799	0.781	1.078	0.966	1.002
pharma	SME	Pre	0.863	0.800	0.710	1.441	1.188	1.245
pharma	non-SME	Post	0.674	0.657	0.633	0.814	0.774	0.765
pharma	non-SME	Pre	0.619	0.604	0.574	0.762	0.725	0.708

Losers-only uses only drop-out bids (upward-biased: misses the lowest cost per auction). All-bidders adds winners as if their final bid were a point observation (downward-biased: winner final bid $\geq$ true cost). Turnbull (1976) NPMLE treats winners as left-censored in $c_{(2)}$ (true upper bound), iterating EM self-consistency until convergence. The empirical ordering is Turnbull $\leq$ all-bidders $\leq$ losers-only at the median in all 8 strata and at the 75th percentile in 6 of 8 strata; the two pharma non-SME cells reverse, with Turnbull marginally below all-bidders by 0.01–0.02 of the reference price.

Table E.22: F_c regime sensitivity: BNE decomposition by sampling rule

Class	Regime	$\bar{p}_{S_1}$	$\bar{p}_{S_2}$	$\bar{p}_{S_3}$	Δ total	% int.	% entry
non-pharma	Turnbull	0.731	1.111	0.978	+0.2462	74.0	26.0
non-pharma	all-bidders	0.759	1.152	1.018	+0.2585	74.5	25.5
non-pharma	losers-only	0.820	1.259	1.096	+0.2753	72.9	27.1
pharma	Turnbull	0.604	1.063	0.962	+0.3573	82.0	18.0
pharma	all-bidders	0.656	1.141	0.964	+0.3079	73.3	26.7
pharma	losers-only	0.692	1.288	1.039	+0.3469	70.6	29.4

Three F_c sampling regimes applied to the same BNE engine (2,000 MC auctions per scenario, UH-clean bids, endogenous entry). Losers-only: ECDF of drop-out bids (biased up). All-bidders: ECDF of winner + loser bids (S4 final; biased up less). Turnbull: NPMLE with winners left-censored at $c_{(2)}$ (point ID under English-reverse IPV). Turnbull is the rigorous benchmark; “all” is the S4 proxy. Small Turnbull–all gap ratifies the S4 headline.

Table E.23: Strict-primitive-invariance benchmark ($F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$)

Class	Δ total	% intensive	Welfare loss (% of p_{S_1})	$w_\star^{\text{SME}}$
non-pharma	+0.29	85	30.0	–
pharma	+0.47	79	39.2	0.7

Strict-invariance benchmark that replaces $F_c^{\text{SME,Post}}$ with $F_c^{\text{SME,Pre}}$ while preserving the observed Post-policy SME entry count. In non-pharma, the 10% price preference remains Pareto-superior. In pharma, the ranking reverses under strict invariance, with indifference weight $w_\star^{\text{SME}} = 0.7$ instead of 2.61 under the main equilibrium-selection specification.

Table E.24: GPV bandwidth robustness: median cost $c_{0.5}$ by h factor

Class	Type	Period	$h/h_{\text{Silverman}}$					range
			0.5	0.75	1.0	1.5	2.0	
non-pharma	SME	Post	0.724	0.724	0.723	0.720	0.716	0.009
non-pharma	SME	Pre	0.717	0.716	0.716	0.712	0.707	0.009
non-pharma	non-SME	Post	0.617	0.615	0.613	0.609	0.604	0.013
non-pharma	non-SME	Pre	0.587	0.585	0.582	0.576	0.570	0.017
pharma	SME	Post	0.866	0.863	0.858	0.849	0.839	0.027
pharma	SME	Pre	0.704	0.699	0.694	0.680	0.668	0.036
pharma	non-SME	Post	0.635	0.632	0.630	0.623	0.617	0.018
pharma	non-SME	Pre	0.625	0.623	0.619	0.612	0.603	0.022

Each column reports $c_{0.5}$ of the GPV-inverted F_c in Convite under bandwidth $h = h_{\text{factor}} \cdot 1.06 \cdot \sigma_b \cdot n^{-1/5}$ (Silverman's rule). Final column is $\max - \min$ across the 5 h values. Robustness means $\text{range} \ll c_{0.5}$ itself (typically $<5\%$ of baseline).

Table E.25: Bootstrap 95% CI for BNE decomposition (B=500, cluster at auction level)

Class	Regime	Δ total [95% CI]	share int. [95% CI]	share entry [95% CI]
non-pharma	losers	0.253 [0.201, 0.309]	74.0 [65.9, 87.9]	26.1 [12.1, 34.1]
non-pharma	all	0.236 [0.186, 0.289]	73.8 [64.9, 88.1]	26.2 [11.9, 35.1]
non-pharma	turnbull	0.253 [0.197, 0.310]	71.7 [64.5, 82.0]	28.4 [18.0, 35.5]
pharma	losers	0.361 [0.294, 0.428]	70.8 [63.1, 86.9]	29.2 [13.1, 36.9]
pharma	all	0.305 [0.247, 0.364]	69.9 [62.5, 82.9]	30.1 [17.1, 37.5]
pharma	turnbull	0.341 [0.271, 0.417]	68.7 [61.5, 82.2]	31.3 [17.8, 38.5]

Cluster bootstrap at the auction level, B=500 replicates. Each replicate resamples auctions with replacement within (period $\times$ pharma) strata, refits F_c in all three regimes, and runs the BNE MC with 500 auctions per scenario. CI endpoints are empirical 2.5 and 97.5 percentiles over the B replicates. Point estimates in main text use the full sample.

Table E.26: Filter sensitivity: BNE decomposition under alternative sample cuts

Class	Filter	n bids (pooled)	$\bar{p}_{S_1}$	$\bar{p}_{S_3}$	Δ total	% int.
non-pharma	baseline	269,521	0.769	1.023	+0.2537	73.3
non-pharma	tight	224,532	0.740	0.956	+0.2153	73.4
non-pharma	very-tight	213,345	0.702	0.877	+0.1755	83.6
non-pharma	loose	270,505	0.767	1.023	+0.2565	71.1
pharma	baseline	269,521	0.647	0.977	+0.3305	73.9
pharma	tight	224,532	0.619	0.922	+0.3031	86.9
pharma	very-tight	213,345	0.591	0.840	+0.2493	95.0
pharma	loose	270,505	0.664	1.016	+0.3515	72.5

Four filter configurations applied to Pregão UH-clean bids: baseline uses $c_\epsilon \leq 3$ and $n \geq 2$; tight cuts $c_\epsilon \leq 2$ with $n \geq 3$; very-tight $c_\epsilon \leq 1.5$ with $n \geq 3$; loose $c_\epsilon \leq 5$. Column “ n bids (pooled)” is the post-filter total bid count across both classes; the BNE simulation runs separately by class on its corresponding subset. Entry counts held fixed across filters. All-bidders regime, BNE with 2,000 MC draws per scenario.

Table E.27: Temporal window sensitivity: BNE decomposition, 18m / 12m / 6m

Class	Window	n auctions (pooled)	$N_{\text{Pre}}^{\text{SME}}$	$N_{\text{Post}}^{\text{SME}}$	Δ total	% int.	% entry
non-pharma	18m	97,993	0.86	1.74	+0.2716	79.7	20.3
non-pharma	12m	64,494	0.96	1.59	+0.2667	75.4	24.6
non-pharma	6m	31,107	1.05	1.50	+0.2688	76.4	23.6
pharma	18m	97,993	0.44	1.08	+0.2920	67.0	33.0
pharma	12m	64,494	0.48	0.90	+0.3096	73.0	27.0
pharma	6m	31,107	0.49	0.84	+0.3375	73.6	26.4

Windows are in Stata monthly-date units centered at cutoff 698 (March 2018): 18m = [680,715], 12m = [686,709], 6m = [692,703]. Column “ n auctions (pooled)” is the post-window total Pregão auction count across both classes; the BNE simulation runs separately by class. Entry counts N_τ^k are recomputed within each window. Cost distributions F_c are re-estimated (all-bidders regime). Narrower windows reduce power; the 6m window may hide steady-state behavior masked by short-run noise. Baseline window is 18m.

Table F.28: Welfare decomposition: allocative DWL + MCPF distortion, by λ

Class	λ	$\bar{p}_{S_1}$	Δ_{gov}	DWL _{alloc}	MCPF dist.	Total loss	% of p_{S_1}
non-pharma	0.15	0.777	0.242	0.150	0.036	0.187	24.01
non-pharma	0.20	0.777	0.242	0.150	0.048	0.199	25.57
non-pharma	0.30	0.777	0.242	0.150	0.072	0.223	28.68
non-pharma	0.40	0.777	0.242	0.150	0.097	0.247	31.79
non-pharma	0.45	0.777	0.242	0.150	0.109	0.259	33.34
pharma	0.15	0.655	0.328	0.210	0.049	0.259	39.49
pharma	0.20	0.655	0.328	0.210	0.066	0.275	41.99
pharma	0.30	0.655	0.328	0.210	0.098	0.308	47.00
pharma	0.40	0.655	0.328	0.210	0.131	0.341	52.00
pharma	0.45	0.655	0.328	0.210	0.147	0.357	54.50

Per-auction welfare decomposition, 5,000 MC draws under all-bidders UH-clean F_c . $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$ is the government's extra payment; $\text{DWL}_{\text{alloc}} = c_{(1)}^{S_3} - c_{(1)}^{S_1}$ is the extra production cost from allocating the auction to a possibly non-optimal SME bidder. MCPF distortion = $\Delta_{\text{gov}} \cdot \lambda$: each additional R\$ of government outlay carries λ in tax distortion. Total loss = DWL + MCPF; reported as % of the baseline p_{S_1} (open, Pre pool). The p_{S_1} and Δ_{gov} values reported here come from an independent 5,000-draw welfare simulation; the 2,000-draw central decomposition (Table D.18) gives $p_{S_1} = 0.759/0.656$ and $\Delta_{\text{gov}} = 0.259/0.308$ in non-pharma/pharma, within Monte Carlo sampling noise of the welfare-simulation values.

Table F.29: Welfare loss 95% CI (B=500 cluster bootstrap)

Class	λ	Δ_{gov} [95% CI]	Total loss as % of p_{S_1} [95% CI]
non-pharma	0.20	0.236 [0.178, 0.294]	25.01 [18.87, 31.44]
non-pharma	0.30	0.236 [0.178, 0.294]	28.09 [21.30, 35.27]
non-pharma	0.40	0.236 [0.178, 0.294]	31.17 [23.80, 39.12]
pharma	0.20	0.307 [0.229, 0.387]	40.82 [31.42, 50.42]
pharma	0.30	0.307 [0.229, 0.387]	45.55 [35.13, 56.18]
pharma	0.40	0.307 [0.229, 0.387]	50.28 [38.59, 62.03]

Cluster bootstrap at auction level, B=500 replicates; each replicate resamples auctions with replacement within (period $\times$ pharma), refits all-bidders F_c , runs a 500-draw welfare MC. CI endpoints are empirical 2.5/97.5 percentiles. Same λ grid as the point estimates.

Table F.30: Welfare ranking stability of V0 (set-aside) vs. V3 (10% preference) across λ

λ	Non-pharma		Pharma		
	Loss(V0)	Ranking	Loss(V0)	Ranking (main)	Ranking (strict-inv)
0.15	24.0%	V3 > V0	38.4%	V3 > V0	V0 > V3
0.20	25.6%	V3 > V0	40.8%	V3 > V0	V0 > V3
0.30	28.7%	V3 > V0	47.0%	V3 > V0	V0 > V3
0.40	31.8%	V3 > V0	50.2%	V3 > V0	V0 > V3
0.45	33.3%	V3 > V0	52.5%	V3 > V0	V0 > V3

Welfare ranking V3 (10% price preference) vs. V0 (full SME-only set-aside) under both the main equilibrium-selection specification and the strict-primitive-invariance benchmark, evaluated across the MCPF grid $\lambda \in \{0.15, 0.20, 0.30, 0.40, 0.45\}$. Loss(V0) is reported in percent of p_{S_1} . “V3 > V0” denotes welfare-dominance of the preference rule over the set-aside; “V0 > V3” is the reversal. The non-pharma ranking is V3 > V0 across the entire λ range under both specifications. The pharma ranking is V3 > V0 across the entire λ range under the main specification but V0 > V3 across the entire λ range under strict invariance. The ranking is therefore not driven by the choice of λ ; it is determined by the equilibrium-selection vs. primitive-invariance treatment of the post-policy SME pool.

Table F.31: Alternative Policy Variants (APV) — counterfactual price effects

Class	$\bar{p}_{S_1}$	V0 (main)	V1 (50%)	V2 (no entry)	V3 (10% pref.)
non-pharma	0.759	+0.2585	+0.0925 [35.8%]	+0.3930 [152.0%]	-0.0036 [-1.4%]
pharma	0.656	+0.3079	+0.1032 [33.5%]	+0.4847 [157.4%]	+0.0015 [0.5%]

Price Δ relative to observed S_1 (open, Pre pool). V0 = full 100% SME set-aside with endogenous Post pool (main text). V1 = partial 50/50 mix. V2 = SME-only but Pre pool (isolates intensive margin). V3 = open auction with 10% price preference for SMEs. Bracketed values are each variant’s Δ as % of V0’s Δ , so V1 = 50% means “half the V0 effect”.

Table F.32: Welfare and SME win-rate across price-preference rates

Class	Preference rate	$\bar{p}$	Δ vs S_1	% of V0 Δ	SME win-rate (%)	Welfare loss (%)
non-pharma	0% pref.	0.772	+0.0103	4.0	17.4	0.49
non-pharma	5% pref.	0.769	+0.0067	2.6	19.8	0.89
non-pharma	10% pref.	0.761	-0.0008	-0.3	21.3	0.11
non-pharma	15% pref.	0.761	-0.0012	-0.5	24.9	0.19
non-pharma	20% pref.	0.775	+0.0131	5.1	27.8	1.68
non-pharma	25% pref.	0.779	+0.0166	6.5	29.5	1.90
non-pharma	30% pref.	0.800	+0.0382	14.9	32.7	4.14
non-pharma	Full set-aside (V0)	1.019	+0.2568	100.0	100.0	28.34
pharma	0% pref.	0.657	+0.0015	0.5	15.3	-0.20
pharma	5% pref.	0.649	-0.0058	-2.0	16.3	-0.32
pharma	10% pref.	0.654	-0.0010	-0.4	19.5	-0.46
pharma	15% pref.	0.659	+0.0035	1.2	19.1	0.79
pharma	20% pref.	0.656	+0.0010	0.4	20.8	1.05
pharma	25% pref.	0.662	+0.0063	2.1	21.9	0.59
pharma	30% pref.	0.674	+0.0191	6.5	23.5	2.67
pharma	Full set-aside (V0)	0.949	+0.2934	100.0	100.0	44.20

Bayes-Nash equilibrium simulation, $B = 3,000$ MC draws per cell. Preference rate k scores SME bids by a factor $(1 - k)$ for winner selection while paying the actual winning bid (Vickrey-equivalent translation). Baseline S_1 is the open regime at the Pre-period pool; V0 is the full set-aside with endogenous Post-period SME pool. Δ is the simulated price effect relative to S_1 , in reference-price units. The nonzero Δ at $k = 0$ reflects Monte Carlo sampling noise relative to the S_1 baseline (true $\Delta = 0$ by construction). Welfare loss = $(DWL_{\text{alloc}} + \lambda \cdot \Delta) / p_{S_1}$, $\lambda = 0.30$. The qualitative pattern: as k rises from 0 to 30 percent, the SME win-rate climbs sharply (from 17.4 to 32.7 percent in non-pharma and from 15.3 to 23.5 percent in pharma) while the welfare cost rises slowly until around 15–20 percent, then accelerates. The 10 percent rate raises SME win rates modestly while keeping welfare loss near zero, in contrast to the full set-aside, which delivers a 100 percent SME win rate at a 28–44 percent welfare loss.

Appendix G. IPV Restriction and Pharmaceutical Procurement

A reader might argue that pharmaceutical procurement involves item-level specifications—ANVISA compliance, brand-name regulatory history, storage chain, production line provenance—that are not well captured by an independent-private-values model with only type-level asymmetry. The concern is that what looks like a bidder-specific private value is in fact a common quality signal that correlates bidder valuations within an auction, in which case Assumption 1 is misspecified and the estimated F_c^k absorbs variation that should be attributed to a common factor.

Three features of the BEC setting address this concern. First, the sample is restricted to Pregão, the electronic reverse-auction format reserved for goods with standardized specifications at the item level; BEC’s CADMAT catalog is designed to strip cross-item quality variation within class. This is a different data-generating environment from the construction contracts studied in [Bajari et al. \(2014\)](#) or the Bayesian auctions of [Hendricks et al. \(2003\)](#), where contract-specific complexity drives correlation. Second, the CMED price ceiling applied to pharmaceutical class 6531 is itself a standardization device: any CMED-regulated drug clears a uniform quality bar, and the residual variation across suppliers is on production cost and distribution scale rather than on quality primitives. Third, the Krasnokutskaya auction-level heterogeneity correction (Assumption 2) allows for an auction-common component a_t that absorbs exactly the kind of unobserved scale or specification variation the objection raises. The Turnbull NPMLE robustness relaxes Assumption 2 entirely and delivers the same decomposition within 16 percent of the total effect across classes—with the largest gap (16 percent) in pharmaceutical Turnbull and a 5 percent gap in non-pharmaceutical Turnbull (Table E.22)—so the main results do not hinge on the Gaussianity of a_t or on the correctness of the BLP shrinkage.

The pharmaceutical welfare-dominance result of Section 9 is therefore subject to two qualifications. First, the strict-invariance check of Section 8 reverses the pharmaceutical ranking: the welfare-dominance result in pharmaceuticals is conditional on the equilibrium-selection treatment of $F_c^{\text{SME,Post}}$. Second, the IPV diagnostic supports Assumption 1 on the available evidence but does not constitute a definitive test. The non-pharmaceutical welfare-dominance result is preserved across both qualifications.

Appendix H. Coincident Regulatory Events at the March 2018 Cutoff

The March 2018 cutoff might in principle coincide with other Brazilian regulatory events that could confound identification—CMED adjustments, platform upgrades, tax changes, federal procurement reform. Each is worth checking explicitly.

CMED list-price adjustments occur annually at the beginning of each year and apply to all CMED-regulated drugs, not to Group 65 specifically; a CMED adjustment would appear in the pre-period Group-65 data as well as in the never-treated product groups of the DiD benchmark (Appendix [Appendix I](#)), and the DiD would difference it out. BEC platform architecture was stable throughout the sample window; the catalog and auction format did not change. Federal Law 14,133/2021 did not exist during the window—the relevant federal framework was the 2006 statute, unchanged throughout. Any Group-65-specific shock that could substitute for the SME rule would need to coincide with March 2018, hit only Group 65, and leave no trace in publicly available administrative records. The DiD pre-trend and placebo diagnostics on the same sample are consistent with the timing being uncorrelated with such shocks; the present design treats the cutoff as plausibly exogenous on those diagnostics rather than as definitively so.

Appendix I. Appendix: Reduced-Form Benchmark

The structural exercise of Sections 6–7 is anchored to a magnitude in the data: the absolute log-price effect of the March 2018 cutoff. This appendix reports that magnitude as a difference-in-differences against the 76 never-treated product groups, on the same sample window used by the structural exercise. The DiD is included for design validation rather than as a competing estimate; the structural exercise interprets the within-auction and participation-margin channels that this number aggregates.

The estimating equation is

$$\log p_{i,j,t}^{\text{final}} = \alpha_i + \mu_t + \beta (\text{g65}_j \cdot \text{Pre}_t) + X'_{i,j,t} \gamma + \varepsilon_{i,j,t}, \quad (\text{I.1})$$

where i indexes items, j indexes product groups, t indexes calendar months, α_i is an item fixed effect, μ_t is a calendar-month fixed effect, $X_{i,j,t}$ collects auction-format and quantity controls, and β identifies the difference between the open-regime price level (Pre , $\text{g65} = 1$) and the post-cutoff level relative to the never-treated control set, in the difference-in-differences-in-reverse sense of [Kim and Lee \(2019\)](#): a negative coefficient on the pre-period dummy is the open-regime discount that the SME-only rule undoes after March 2018.

Table [I.33](#) reports $\hat{\beta}$ across the 6-, 12-, and 18-month windows on each side of the cutoff. The 18-month specification (the structural baseline) gives $\hat{\beta} = -0.109$ without and -0.113 with PBU controls, both significant at the 1 percent level on standard errors clustered at the item level. The interpretation is that absolute winning prices were about 11 percent lower under the open regime than they were after the SME-only extension. The 12-month and 6-month windows give similar estimates (-0.108 and -0.142 respectively); the 6-month coefficient captures the immediate post-cutoff price response before the participation margin has equilibrated, while the 12-month and 18-month coefficients

Table I.33: Prices (log): Pre-switch-period effect on group 65

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	-0.1417*** (0.0146)	-0.1482*** (0.0144)	-0.1079*** (0.0133)	-0.1077*** (0.0129)	-0.1087*** (0.0120)	-0.1133*** (0.0117)
Sealed bids	-0.1409*** (0.0086)	-0.2176*** (0.0130)	-0.1380*** (0.0077)	-0.1991*** (0.0122)	-0.1455*** (0.0077)	-0.2018*** (0.0114)
lquantity	-0.2616*** (0.0082)	-0.2986*** (0.0089)	-0.2634*** (0.0075)	-0.2974*** (0.0080)	-0.2617*** (0.0073)	-0.2949*** (0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1919	0.2132	0.1943	0.2120	0.1947	0.2118
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

reflect progressive absorption of the within-auction shock by SME entry, consistent with the structural-window reading in Section 8. The 10–11 percent magnitude that anchors the structural calibration is therefore the 18-month log-price coefficient.

Table I.34: Price Efficiency: Final Price / Reference Price (%)

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	-0.0563*** (0.0118)	-0.0596*** (0.0113)	-0.4259 (0.3813)	-0.4723 (0.4260)	-0.2921 (0.2623)	-0.3237 (0.2949)
Sealed bids	-0.0622*** (0.0027)	-0.0498*** (0.0044)	0.0279 (0.0850)	0.0926 (0.1385)	-0.0057 (0.0531)	0.0330 (0.0835)
lquantity	-0.0151*** (0.0018)	-0.0157*** (0.0018)	0.1226 (0.1381)	0.1351 (0.1496)	0.0717 (0.0876)	0.0777 (0.0939)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0037	0.0023	0.0001	0.0001	0.0000	0.0000
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

Table I.34 reports the same estimating equation with the price-to-reference ratio $p^{\text{final}}/p^{\text{ref}}$ as the outcome on a level (not log) scale, so coefficients are interpretable as

percentage-point shifts in the ratio. The 6-month window gives -0.060 (with PBU controls), consistent with the 18-month log-price effect of -0.11 once the average ratio of about 0.65 in the pre-period is taken into account. The 12-month and 18-month coefficients on this level outcome are smaller in magnitude and statistically insignificant, reflecting that the price-to-reference ratio is bounded above by the buyer’s reservation level and therefore averages over a compressed support; the log-price specification of Table I.33 is the headline. The price-to-reference outcome is reported as a robustness check on the headline log-price magnitude rather than as a structural input.

Appendix I.1. Event-study and placebo diagnostics

The DiD specification is reinforced by an event-study around the March 2018 cutoff and three placebo treatment dates. Figure I.6 plots $\hat{\beta}_s$ for six semesters around the cutoff (each semester is six calendar months) with item and semester fixed effects, on the same Group-65 vs. 76-control panel used by Table I.33; the semester immediately preceding the cutoff is the omitted reference. The semester binning aggregates the monthly noise that would otherwise dominate the visual reading. Pre-cutoff coefficients are small in magnitude relative to the post-cutoff coefficients; the post-cutoff coefficients are stably negative and consistent with the -0.10 to -0.11 DiD coefficient. The placebo evidence in Table I.35 adds discipline at fake cutoff dates and is read jointly with the event-study below.

Table I.35: Placebo Tests: Fake Treatment Dates

	Log prices			Log firms			Log bids			Distance		
	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17
$g65 \times Pre_{placebo}$	-0.0129 (0.0182)	-0.0297** (0.0132)	-0.0342*** (0.0128)	-0.1130*** (0.0074)	-0.0764*** (0.0081)	-0.0440*** (0.0082)	-0.1574*** (0.0107)	-0.0434*** (0.0124)	-0.0826*** (0.0116)	3.5995 (2.6235)	-3.6015 (3.1194)	6.2414** (2.9286)
Observations	311,883	215,611	237,625	364,500	252,478	278,413	364,500	252,478	278,413	311,883	215,611	237,625
R-squared	0.1964	0.1972	0.1924	0.2179	0.2262	0.2200	0.2245	0.2318	0.2336	0.0007	0.0005	0.0007
Item + Month FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using three fake treatment dates (Sep 2017, Mar 2017, Jun 2017) on pre-treatment data only. Sep 2017 uses window [680, 697]; Mar 2017 uses [680, 691]; Jun 2017 uses [683, 695]. Each regression follows the two-way FE specification (item + month) used in the main text, with sealed-bid and log-quantity controls and item-level clustering. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table I.35 reports placebo treatment dates at September 2017, March 2017, and June

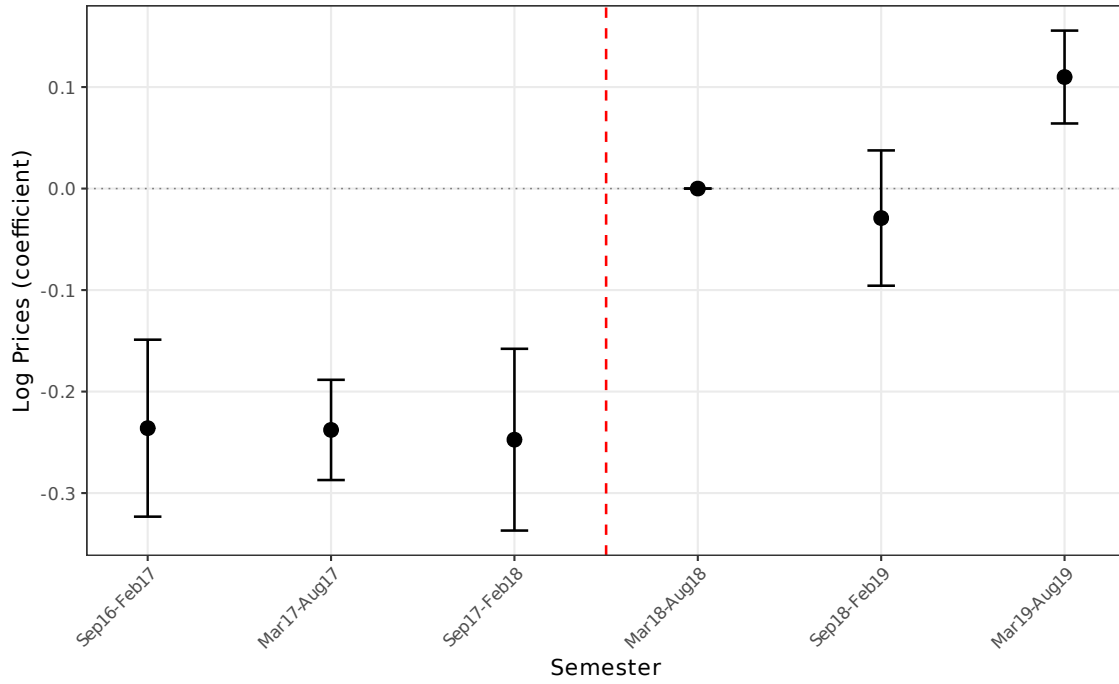


Figure I.6: *Event-study coefficients $\hat{\beta}_s$ on $\log p^{\text{final}}$ around the March 2018 cutoff, semester aggregation.* Coefficients from a within-item difference-in-differences specification with item and semester fixed effects, on Group 65 vs. the 76 never-treated product groups. Time is binned into six six-month semesters around the cutoff; the semester immediately preceding the cutoff is the omitted reference (zero by construction). Standard errors clustered at the item level; whiskers show 95% confidence intervals. The post-cutoff semesters display coefficients that align in direction and magnitude with the -0.10 to -0.11 headline DiD coefficient.

2017. The price coefficients for the three placebos are -0.013 (statistically insignificant), -0.030 (significant at the 5 percent level), and -0.034 (significant at the 1 percent level). The placebo evidence does not establish absence of pre-period correlation in Group-65 prices: low-frequency variation in the treated group is detectable at fake cutoffs near the real one. The relevant test for whether the DiD coefficient anchors a structural input is therefore not absence of pre-trend, but magnitude separation. The actual March 2018 coefficient of -0.108 to -0.142 is three to eight times larger in absolute value than any of the three placebo coefficients and is significant at the 1 percent level in every specification. The structural calibration is anchored to a treatment effect of demonstrably larger magnitude than any pre-period drift the placebo design can detect; the residual concern that some of the real-cutoff coefficient reflects gradual anticipation rather than discrete treatment is a bound, not a refutation, on the structural input. Recent guidance on event-study contamination ([Borusyak et al., 2024](#); [Sun and Abraham, 2021](#)) emphasizes that staggered or anticipated treatments produce coefficients that combine the discrete-treatment effect with anticipation dynamics; the magnitude-separation defense above bounds the anticipation contribution at the difference between the real-cutoff and placebo coefficients, which is at least 67 percent of the real-cutoff magnitude. Read jointly, the event-study and the placebo table support the interpretation of the DiD coefficient as a treatment effect with at most moderate anticipation contamination.

Appendix I.2. Bridge to the structural counterfactual

The structural exercise of the main text recovers the within-auction and participation-margin decomposition of the -0.11 DiD coefficient. Section 6 translates the structural counterfactual into normalized units: $p_{S_3} - p_{S_1} = +0.259$ in non-pharma and $+0.308$ in pharma, of which the within-auction component accounts for 74.5 and 73.3 percent respectively. Expressed as a ratio to the open-regime baseline p_{S_1} , these are $0.259/0.759 \approx 34$ percent in non-pharma and $0.308/0.656 \approx 47$ percent in pharma; the corresponding

DiD-implied price ratio is $\exp(0.11) - 1 \approx 12$ percent on absolute prices. The structural counterfactual is therefore three to four times the DiD coefficient, not coincident with it. The two exercises measure related but distinct objects: the DiD recovers the average log-price change in the treated set conditional on items completing, with item and month fixed effects absorbing the cross-sectional cost variation; the structural counterfactual recovers the simulated mean of the second-order statistic $c_{(2)}$ under the BNE-equivalent transaction price, on the cleaned cost residuals after the Krasnokutskaya auction-level scale is shrunk out, conditional on the bidder pool that endogenously enters under each regime. The DiD averages over the realized price distribution including item- and PBU-level shocks that the structural exercise normalizes away; the structural counterfactual averages over a counterfactual primitive distribution that the DiD never observes. The DiD therefore identifies the magnitude of the policy effect on a directly observable outcome; the structural exercise identifies the within-auction vs participation-margin decomposition of a transformed counterfactual object. The DiD is reported here for design validation; the gap between the two magnitudes is mechanical, not contradictory, and the welfare arithmetic of Section 7 is anchored to the structural object on the structural normalization.